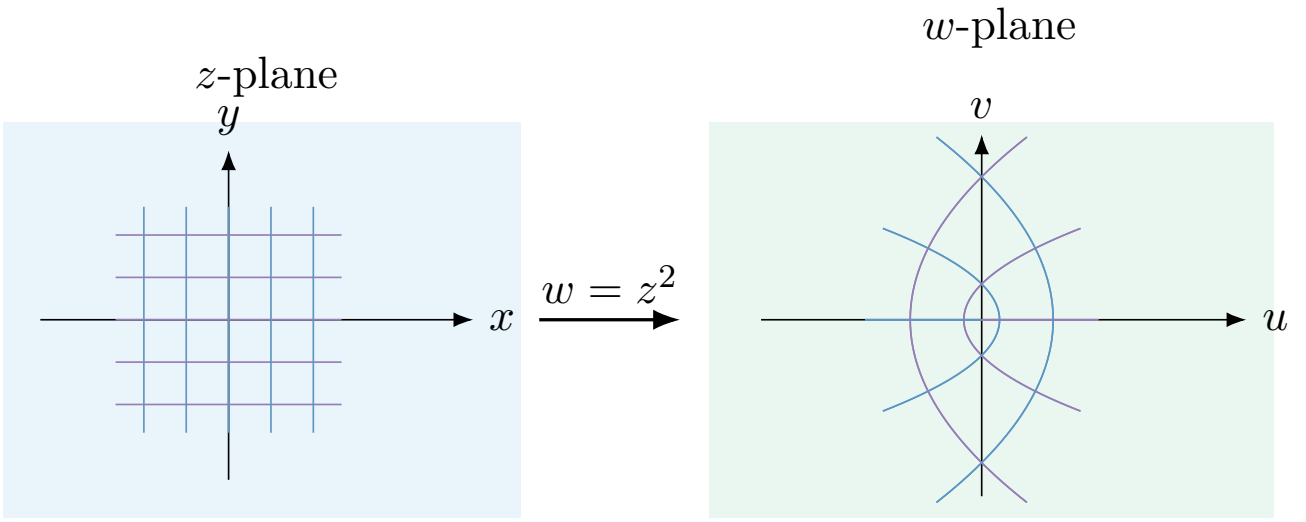
COMPLEX FUNCTIONS

Complete Learning Notes

Geometry, holomorphic structure, and contour methods

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Organized by Shang Wu



Definition: The learning route

Identify the domain → inspect analyticity and topology → choose a representation → derive → verify.

Detailed proofs and derivations, worked examples, geometric figures, standard solution workflows, and a mind map after every learning chapter.

Based on the supplied current-semester syllabus and Marco Farinati, *GTIIT–Winter 2025/2026–Function Theory–Preliminar Lecture Notes*, 142-page edition. Chapters 1–14 cover the syllabus; Chapter 15 is optional enrichment. Explanations, diagrams, and practice are newly prepared. No past examination was supplied.

Course Map and Prerequisites

Chapters 1–14 follow the supplied current-semester syllabus through real integrals by residues. Chapter 15 is optional enrichment. The source is Marco Farinati's 142-page Function Theory lecture file, labeled Winter 2025/2026; printed and PDF page numbers agree. No past examination was supplied.

Ch.	Knowledge and central skill	Source pages	Practice IDs
1	Complex arithmetic, geometry, matrix model	4–7	CF1.1–CF1.6
2	Topology, sequences, exponential and roots	8–13	CF2.1–CF2.6
3	Complex derivative, real differential, C–R	14–20	CF3.1–CF3.6
4	Local inverses, logarithms, branch domains	20–22	CF4.1–CF4.6
5	Harmonic conjugates, fluids, conformality	22–29	CF5.1–CF5.6
6	Generalized circles, Möbius maps, sphere	30–43	CF6.1–CF6.6
7	Series tests, radii, boundary convergence	44–60	CF7.1–CF7.6
8	Pointwise/uniform/local uniform limits	61–64; 75; 80–84	CF8.1–CF8.6
9	Contours, primitives, topology, Cauchy's theorem	65–76	CF9.1–CF9.6
10	Cauchy formulas, estimates, rigidity	77–90	CF10.1–CF10.6
11	Taylor series, zero orders, identity theorem	90–97	CF11.1–CF11.6
12	Laurent series, singularities, residues	97–116	CF12.1–CF12.6
13	Argument principle, Rouché, infinity	95–97; 116–122	CF13.1–CF13.6
14	Real integral contours and arc estimates	123–132	CF14.1–CF14.6
15	Optional: Schwarz, disk maps, Riemann mapping	133–142	CF15.1–CF15.6

Error Check: What the evidence does and does not say

Every supplied syllabus topic is mapped above. Geometry, convergence regions, contour selection, theorem hypotheses, and full proofs are training priorities supported by the lecture exercises, not measured examination frequencies. The practice set contains newly written course-aligned questions, not reconstructed school examinations. Advanced disk-mapping results are not asserted to be examination requirements.

Notation and prerequisite bridge

Write $z = x + \mathrm{i}y$, $\bar{z} = x - \mathrm{i}y$, $\mathbb{D} = \{|z| < 1\}$, and $\mathbb{H} = \{\operatorname{Im} z > 0\}$. A *domain* is a nonempty connected open set. Positive boundaries are counterclockwise outside and clockwise around holes; contours are piecewise C^1. Holomorphy requires a complex derivative at every point of an open domain, not merely at one point.

Prerequisites: real-linear maps and matrices (Algebra 2M); multivariable derivatives, Green's theorem, and series (Calculus); harmonic functions (PDE). Complex-potential applications state their physical sign convention explicitly.

Standard Workflow: An assessable standard solution

State the domain, branches, singularities, and orientation; justify the method; derive the result; handle exceptional cases; check the answer. "Apply residues" names a method but does not supply a worked solution.

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1 Complex Numbers, Algebra, and Plane Geometry

Source map: Farinati, Chapter 1, pp. 4–7; polar geometry from Chapter 2.

1.1 A field built from pairs of real numbers

The aim is not merely to write $\sqrt{-1}$. We want a consistent algebra in which quadratic equations, rotations, and linear maps can be treated together. Define $\mathbb{C} = \mathbb{R}^2$ with

$$(a,b)+(c,d)=(a+c,b+d), \qquad (a,b)(c,d)=(ac-bd,ad+bc).$$

Writing $(a,b) = a + \mathrm{i}b$ gives $\mathrm{i}^2 = -1$. Equality means equality of both real and imaginary parts. The conjugate and modulus are $\bar{z} = x - \mathrm{i}y$ and $|z| = \sqrt{x^2+y^2}$. Direct multiplication gives

$$z\bar{z} = |z|^2, \quad \overline{zw} = \bar{z}\bar{w}, \quad |zw| = |z||w|, \quad z^{-1} = \frac{\bar{z}}{|z|^2} \quad (z \neq 0).$$

The modulus product follows by squaring both nonnegative sides: $|zw|^2 = zw\bar{z}\bar{w} = |z|^2|w|^2$. Division by a complex number means multiplying numerator and denominator by its conjugate, not dividing real and imaginary parts separately.

Worked Example: A reproducible quotient calculation

Find $(3+2\mathrm{i})/(1-\mathrm{i})$. Multiply by $(1+\mathrm{i})/(1+\mathrm{i})$:

$$\frac{(3+2\mathrm{i})(1+\mathrm{i})}{2} = \frac{1+5\mathrm{i}}{2}.$$

Multiplying this result by $1-\mathrm{i}$ recovers $3+2\mathrm{i}$. Its modulus is $\sqrt{26}/2$, agreeing with $\sqrt{13}/\sqrt{2}$.

1.2 The matrix model: the Algebra 2M connection

Multiplication by $z = a + \mathrm{i}b$ is the real-linear map

$$M_z = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = aI + bJ, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I.$$

Computing the product gives $M_z M_w = M_{zw}$. This proves associativity of complex multiplication from associativity of matrix multiplication. Also

$$\det M_z = a^2 + b^2 = |z|^2, \quad M_z^T M_z = |z|^2 I, \quad M_z^{-1} = M_{1/z} \ (z \neq 0).$$

If $z = r\mathrm{e}^{\mathrm{i}\theta}$, then $M_z = rR_\theta$. Nonzero multiplication preserves orientation, scales lengths by r, areas by r^2, and rotates angles by θ. Conjugation, in contrast, has matrix $\mathrm{diag}(1,-1)$, determinant -1, and reverses orientation.

Worked Example: Recognizing a complex-linear real matrix

The matrix $A = \left(\begin{smallmatrix} 2 & -3 \\ 3 & 2 \end{smallmatrix}\right)$ is multiplication by $2+3\mathrm{i}$. Its scale is $\sqrt{13}$ and its angle is $\arctan(3/2)$. The matrix $B = \mathrm{diag}(2,3)$ is not multiplication by any complex number: the diagonal entries would have to coincide. It stretches two perpendicular directions unequally.

1.3 Polar form, arguments, and every root

For $z \neq 0$, $z = r(\cos\theta + \mathrm{i}\sin\theta) = r\mathrm{e}^{\mathrm{i}\theta}$, with $r > 0$. An argument is determined modulo 2π. The principal argument convention used here is $-\pi < \operatorname{Arg} z \leq \pi$ as a pointwise convention; the principal *holomorphic logarithm* later excludes the negative real axis entirely.

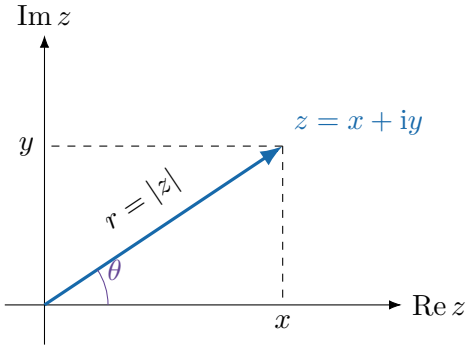


Figure 1.1: Cartesian coordinates resolve addition, while radius and angle resolve multiplication.

From trigonometric angle addition,

$$(r\mathrm{e}^{\mathrm{i}\theta})(s\mathrm{e}^{\mathrm{i}\phi}) = rs\mathrm{e}^{\mathrm{i}(\theta+\phi)}, \qquad (r\mathrm{e}^{\mathrm{i}\theta})^n = r^n\mathrm{e}^{\mathrm{i}n\theta}.$$

For $w = \rho\mathrm{e}^{\mathrm{i}\phi} \neq 0$, solving $z^n = w$ requires $r^n = \rho$ and $n\theta = \phi + 2\pi k$. Hence the distinct roots are

$$z_k = \rho^{1/n}\mathrm{e}^{\mathrm{i}(\phi+2\pi k)/n}, \quad k = 0, \ldots, n-1.$$

There are exactly n: larger integer indices repeat these values, and different listed indices cannot differ by an integer multiple of 2π in angle.

Worked Example: Four roots, with no principal-root shortcut

For $z^4 = -4 = 4\mathrm{e}^{\mathrm{i}\pi}$, the radius is $\sqrt{2}$ and the angles are $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$. The roots are $1+\mathrm{i}, -1+\mathrm{i}, -1-\mathrm{i}, 1-\mathrm{i}$. Each has fourth power -4; specifying only $1+\mathrm{i}$ would discard three solutions.

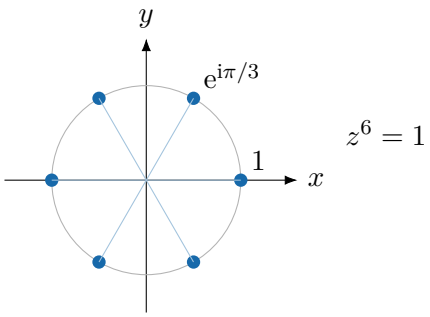


Figure 1.2: Roots of unity are equally spaced; their vector sum is zero when there is more than one root.

1.4 Distances, inequalities, and geometric loci

The identities

$$|z-w|^2 = (z-w)(\bar{z}-\bar{w}), \qquad |z+w|^2 = |z|^2 + |w|^2 + 2\operatorname{Re}(z\bar{w})$$

give the triangle inequality because $\operatorname{Re}(z\bar{w}) \leq |z||w|$. Applying it twice yields $\big||z|-|w|\big| \leq |z-w|$. Equality in $|z+w| \leq |z| + |w|$ occurs when one vector is zero or they point along the same ray.

Worked Example: An Apollonius circle including the exceptional parameter

Solve $|z| = \lambda|z-1|$ for real $\lambda \geq 0$. At $\lambda = 0$, only $z = 0$ works. At $\lambda = 1$, squaring gives $x = 1/2$. Otherwise

$$(1-\lambda^2)(x^2+y^2) + 2\lambda^2 x - \lambda^2 = 0,$$

so the center is $\lambda^2/(\lambda^2-1)$ on the real axis, and the radius is $\lambda/|1-\lambda^2|$. For $\lambda = 2$ the center is $4/3$, radius $2/3$. The original equality, not a rough sketch, decides the geometry.

Standard Workflow: Arithmetic and geometry

Use Cartesian form for linear equations and squared-distance loci. Use polar form for products, powers, and all roots. For a matrix question compare against $aI + bJ$. Separate zero denominators and parameters before completing a square or dividing. Check the final objects in the original equation.

1.5 Chapter Mind Map

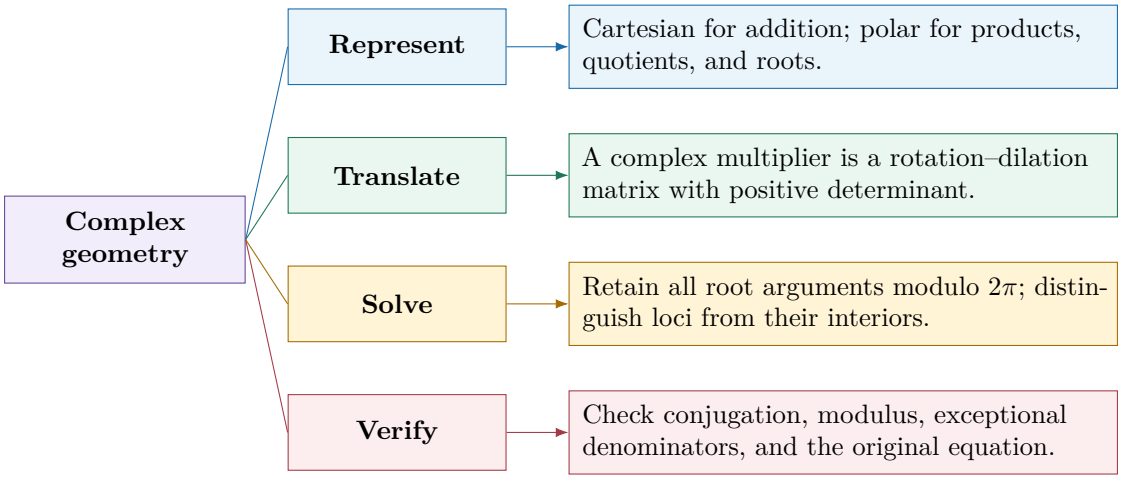


Figure 1.3: Concepts, calculation route, conditions, and checks.

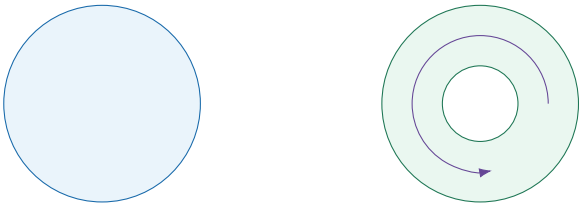
2 Topology, Sequences, and the Complex Exponential

Source map: Farinati, Chapter 2, pp. 8–13; completeness from Chapter 7.

2.1 Domains are part of the problem

A disk $D(a,r) = \{|z-a| < r\}$ is open. A set is open if every point has a disk contained in the set. A domain is open and connected; in the plane, open connected sets are path connected. A compact subset of $\mathbb{C}$ is exactly a closed bounded subset. Compactness lets a continuous modulus attain a maximum and gives a positive distance from a compact contour to a disjoint closed singular set.

A domain is simply connected if every closed continuous loop contracts to a point within it. A disk and a half-plane are simply connected. An annulus and $\mathbb{C} \setminus \{0\}$ are not: a loop surrounding the missing point cannot contract without crossing it. A slit plane is different from a punctured plane; removing a ray can eliminate the obstruction to choosing an argument.



disk: no hole annulus: a loop surrounds a hole

Figure 2.1: Connectedness allows travel between points; simple connectivity additionally rules out noncontractible loops.

Worked Example: The domain determines what may be concluded

The function $1/z$ is holomorphic on both $\mathbb{C}\setminus\{0\}$ and the right half-plane. It has a holomorphic primitive on the half-plane, but none on the punctured plane, because the unit-circle integral equals $2\pi\mathrm{i}$. A local differential equation does not erase a global hole.

2.2 Complex limits reduce to two real limits

For $z_n = x_n + \mathrm{i}y_n$,

$$z_n \to x + \mathrm{i}y \iff x_n \to x \text{ and } y_n \to y.$$

Indeed $|x_n - x|, |y_n - y| \le |z_n - z|$, while $|z_n - z| \le |x_n - x| + |y_n - y|$. The same estimates prove that a complex sequence is Cauchy exactly when both component sequences are Cauchy. Completeness of $\mathbb{R}$ therefore proves completeness of $\mathbb{C}$.

Worked Example: Quantitative convergence, not just a plausible limit

For $z_n = (1+\mathrm{i}/n)/(2-\mathrm{i}/n)$, subtract $1/2$:

$$z_n - \tfrac{1}{2} = \frac{3\mathrm{i}/n}{2(2-\mathrm{i}/n)}, \qquad |z_n - \tfrac{1}{2}| \le \frac{3}{4n}.$$

Thus $n > 3/(4\varepsilon)$ ensures an error below ε. The denominator is bounded away from zero because $|2 - \mathrm{i}/n| \ge 2$.

Worked Example: Bounded does not mean convergent

$z_n = \mathrm{i}^n$ lies on the unit circle but has four constant subsequences with distinct limits. It is neither convergent nor Cauchy. In contrast, $(1+\mathrm{i})/n \to 0$. Always distinguish boundedness, existence of subsequential limits, and a unique full-sequence limit.

2.3 Define the exponential and derive its laws

Define $\exp z = \sum_{n\ge 0} z^n/n!$. Absolute convergence follows from the real series $\sum |z|^n/n!$ (or its ratio $|z|/(n+1) \to 0$). Thus we can regroup even and odd powers to obtain

$$\mathrm{e}^{\mathrm{i}t} = \sum_{k\ge 0} \frac{(-1)^k t^{2k}}{(2k)!} + \mathrm{i}\sum_{k\ge 0} \frac{(-1)^k t^{2k+1}}{(2k+1)!} = \cos t + \mathrm{i}\sin t.$$

The absolutely convergent Cauchy product and the binomial theorem give

$$\mathrm{e}^z\mathrm{e}^w = \sum_{n\ge 0}\sum_{j=0}^{n} \frac{z^j w^{n-j}}{j!(n-j)!} = \sum_{n\ge 0} \frac{(z+w)^n}{n!} = \mathrm{e}^{z+w}.$$

Consequently $\mathrm{e}^z\mathrm{e}^{-z} = 1$, so e^z never vanishes. With $z = x + \mathrm{i}y$, $\mathrm{e}^z = \mathrm{e}^x(\cos y + \mathrm{i}\sin y)$ and $|\mathrm{e}^z| = \mathrm{e}^x$. Also

$$\mathrm{e}^z = \mathrm{e}^w \iff z - w \in 2\pi\mathrm{i}\mathbb{Z}.$$

To prove the reverse implication to periodicity, $e^{z-w} = 1$ forces the real part of $z - w$ to be zero and its imaginary part to be an integer multiple of 2π.

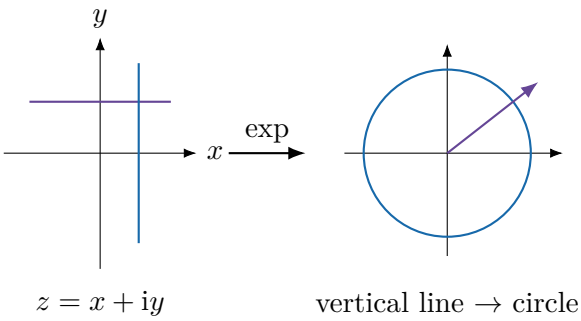


Figure 2.2: Horizontal lines map to rays, while vertical lines map to circles. Periodicity prevents global injectivity.

Worked Example: Solve the complete exponential equation

$e^z = -2i$ implies $e^x = 2$ and $y = -\pi/2 + 2\pi k$. Hence $z = \log 2 + i(-\pi/2 + 2\pi k)$, $k \in \mathbb{Z}$. The real part is unique; the imaginary parts form an infinite arithmetic progression.

2.4 Iteration as a convergence laboratory

The source's Mandelbrot example studies $z_0 = 0$, $z_{n+1} = z_n^2 + c$. At $c = 0$, the sequence is zero. At $c = -1$, it alternates between 0 and -1: bounded, but not convergent. At $c = 1$, it begins $0, 1, 2, 5, 26$, then escapes.

Here is a justified escape criterion. If $|c| \le 2$ and some $r_N = |z_N| > 2$, then $r_{n+1} \ge r_n^2 - 2 > r_n$. If the increasing tail had a finite limit $r > 2$, passing to the inequality would give $r \ge r^2 - 2$, impossible. Hence it diverges in modulus. If $|c| > 2$, start with $r_1 = |c|$ and use $r_{n+1} \ge r_n^2 - |c| \ge r_n(r_n - 1)$ whenever $r_n \ge |c|$ to get escape. The Mandelbrot set consists of parameters giving bounded orbits, not only convergent ones.

Standard Workflow: Limits and elementary functions

Record the domain; estimate a difference or use component limits; keep denominators uniformly nonzero; for exponentials compare moduli and arguments separately; for an iteration prove boundedness or escape rather than relying on a few computed terms.

2.5 Chapter Mind Map

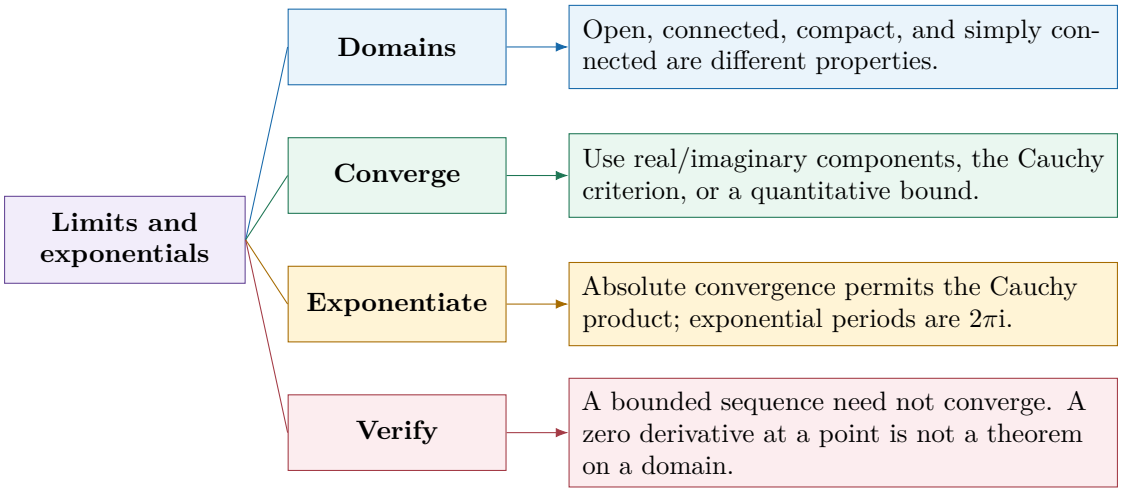


Figure 2.3: Concepts, calculation route, conditions, and checks.

3 Complex Derivatives and Cauchy–Riemann

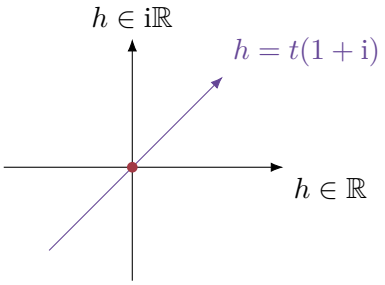
Source map: Farinati, Chapter 3, pp. 14–20.

3.1 One complex derivative controls every direction

For an interior point a, define

$$f'(a) = \lim_{h\to 0,\ h\in\mathbb{C}\setminus\{0\}} \frac{f(a+h)-f(a)}{h}.$$

Equivalently, $f(a+h) = f(a) + Ah + o(|h|)$ for one complex number A. This is much more restrictive than two independent real directional derivatives: the same A multiplies every direction.



Every approach direction must produce the same difference-quotient limit.

Figure 3.1: Agreement along two axes is necessary, but the full limit must also agree along oblique and curved approaches.

Worked Example: A derivative at one point without holomorphy nearby

For $f(z) = |z|^2$, at zero the quotient is $|h|^2/h = \bar{h} \to 0$, so $f'(0) = 0$. At $a \neq 0$, along real increments the quotient tends to $2\operatorname{Re} a$, while along imaginary increments it tends to $-2\mathrm{i}\operatorname{Im} a$; these agree only at zero. Thus f is differentiable exactly at zero and is holomorphic on no nonempty open set.

3.2 Derive the C–R conditions and their exact converse

Write $f = u + \mathrm{i}v$. Real increments give $f' = u_x + \mathrm{i}v_x$, and imaginary increments give $f' = v_y - \mathrm{i}u_y$. Equality yields

$$u_x = v_y, \qquad u_y = -v_x, \qquad f' = u_x + \mathrm{i}v_x = v_y - \mathrm{i}u_y.$$

The correct sufficiency statement includes real differentiability.

Key Result: The real-differential characterization

f is complex differentiable at a if and only if it is real differentiable there and satisfies C–R there. In particular, continuous first partial derivatives in a neighborhood plus C–R at a suffice.

To prove sufficiency, real differentiability gives

$$f(a+h) - f(a) = Df(a)(h) + o(|h|).$$

C–R makes $Df(a) = \left(\begin{smallmatrix} u_x & -v_x \\ v_x & u_x \end{smallmatrix}\right)$, which is multiplication by $A = u_x + \mathrm{i}v_x$. Dividing the remainder by h now gives a term of modulus $o(|h|)/|h| \to 0$. Necessity follows directly from the expansion $Ah + o(|h|)$; it already is a real-differentiability expansion.

The Algebra 2M formulation is $Df(a)J = JDf(a)$, with J from Chapter 1. Solving this commutation equation forces the matrix to be $aI + bJ$. At a nonzero derivative, its determinant is $|f'|^2 > 0$; holomorphic maps do not reflect orientation.

Worked Example: C–R alone at one point can fail

Set $f(0) = 0$ and $f(x+\mathrm{i}y) = xy/\sqrt{x^2+y^2}$ otherwise, with imaginary part zero. All four coordinate partials at zero are zero, so C–R holds. But along $h = t > 0$ the quotient is zero; along $h = t(1+\mathrm{i})$, $t > 0$, it is $1/[\sqrt{2}(1+\mathrm{i})]$. Hence no complex derivative exists. The missing hypothesis is real differentiability, not continuity: $|f(z)| \le |z|/2$ proves continuity at zero.

3.3 Efficient tests and derivative rules

If f is C^1 as a real map, introduce

$$\partial_z = \tfrac{1}{2}(\partial_x - \mathrm{i}\partial_y), \qquad \partial_{\bar{z}} = \tfrac{1}{2}(\partial_x + \mathrm{i}\partial_y).$$

Then $\partial_{\bar{z}} f = 0$ is exactly C–R, and $f' = \partial_z f$. This is a useful notation, not permission to regard z and $\bar{z}$ as unconstrained independent variables in arbitrary limits.

Worked Example: A real-linear family

Every real-linear map has the form $T(z) = az + b\bar{z}$: express $x = (z+\bar{z})/2$, $y = (z-\bar{z})/(2\mathrm{i})$ in its real formula. Its difference quotient is $a + b\bar{h}/h$. Real increments give $a+b$; imaginary increments give $a-b$. Therefore complex differentiability at even one point requires $b = 0$, and then $T' = a$ everywhere.

The usual sum, product, reciprocal, and chain rules follow by substituting differentiability expansions. For the product,

$$(f+\delta f)(g+\delta g) - fg = g\delta f + f\delta g + \delta f \delta g,$$

and the last term is $O(|h|^2)$. For the chain rule, compose the real differentials or substitute the two expansions to obtain $(f\circ g)'(a) = f'(g(a))g'(a)$. The quotient rule requires a nonzero denominator on a neighborhood.

Worked Example: Differentiate only after specifying the domain

For $f(z) = \mathrm{e}^{z^2}/(z-\mathrm{i})$, the domain is $\mathbb{C}\setminus\{\mathrm{i}\}$ and

$$f'(z) = \frac{\mathrm{e}^{z^2}[2z(z-\mathrm{i})-1]}{(z-\mathrm{i})^2}.$$

No derivative at i is claimed. The exponential is entire, and the quotient is holomorphic away from that point.

3.4 Rigidity already visible from C–R

If a holomorphic $f = u + \mathrm{i}v$ takes only real values on a connected domain, then $v = 0$, so C–R gives $u_x = u_y = 0$. Thus u is locally constant and therefore constant throughout the connected domain. The same holds for purely imaginary values. If $|f|$ is constant, differentiate u^2+v^2 and use C–R: at a nonzero value the resulting invertible two-by-two system forces all first partials to vanish. If the modulus is zero, the conclusion is immediate.

Standard Workflow: Differentiability questions

Start with the real formula and its smoothness set. Compute C–R; at smooth points this gives necessity and sufficiency. At exceptional points return to the actual quotient or establish real differentiability. Distinguish the set of differentiability points from an open holomorphy domain. For a counterexample, verify both the advertised hypothesis and the failed conclusion.

3.5 Chapter Mind Map

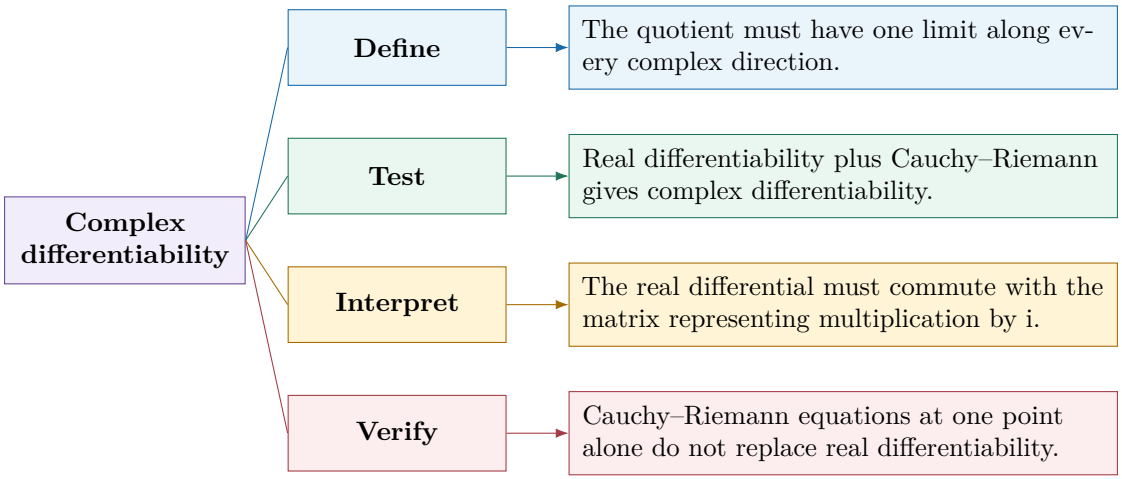


Figure 3.2: Concepts, calculation route, conditions, and checks.

4 Local Inverses, Logarithms, and Branch Choices

Source map: Farinati, Chapter 4, pp. 20–22; logarithm exercises in 3.5; mapping applications in Chapter 15.

4.1 Elementary holomorphic functions beyond the exponential

Polynomials are entire; rational functions are holomorphic where their denominator is nonzero after accounting for removable cancellations. The exponential definitions

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \qquad \cos z = \frac{e^{iz} + e^{-iz}}{2}$$

make sine and cosine entire, with derivatives $\cos z$ and $-\sin z$. Multiplying the exponential expressions proves $\sin^2 z + \cos^2 z = 1$ for all complex z. Expanding $z = x + iy$ gives

$$\sin z = \sin x \cosh y + i \cos x \sinh y, \quad \cos z = \cos x \cosh y - i \sin x \sinh y.$$

Thus complex sine and cosine are not globally bounded: $\sin(iy) = i \sinh y$. Solving their exponential equations gives all zeros: $\sin z = 0$ exactly at $n\pi$, and $\cos z = 0$ exactly at $\pi/2 + n\pi$, $n \in \mathbb{Z}$. They are simple because the derivative at each listed point is nonzero. Consequently $\tan z = \sin z / \cos z$ is meromorphic with simple poles at the cosine zeros, and is holomorphic elsewhere.

Worked Example: An elementary equation needs both real and imaginary parts

Solve $\sin z = 2$. Its imaginary part requires $\cos x \sinh y = 0$. If $y = 0$, the real sine cannot equal two. Hence $\cos x = 0$. For $x = \pi/2 + 2\pi n$, the real equation is $\cosh y = 2$, giving $y = \pm \log(2 + \sqrt{3})$. At $x = 3\pi/2 + 2\pi n$ it would require $-\cosh y = 2$, impossible. These are all solutions.

4.2 Local invertibility is not global injectivity

If f is holomorphic and $f'(a) \neq 0$, the real Jacobian has determinant $|f'(a)|^2 > 0$. The real inverse theorem gives a local inverse, whose differential is multiplication by $1/f'(a)$. Thus the inverse is holomorphic and

$$(f^{-1})'(f(a)) = 1/f'(a).$$

At this point one may assume f is C^1 to apply the familiar real theorem; Chapter 11 proves the automatic smoothness of every holomorphic function. Nonconstant holomorphic maps are open even at critical points, but that stronger theorem will be proved after zero counting.

Worked Example: A nonzero derivative does not prevent repeated values

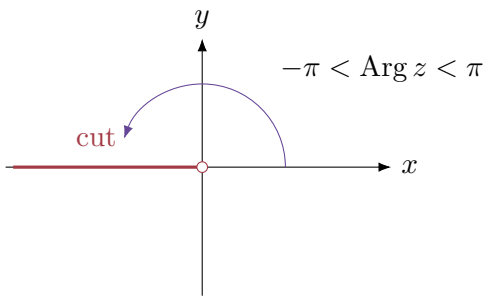
$f(z) = \mathrm{e}^z$ has nonzero derivative everywhere, but $f(z + 2\pi\mathrm{i}) = f(z)$. Restricting to a strip of imaginary width 2π removes this repetition and yields a logarithm on a slit plane. By contrast, z^2 has derivative zero at zero and is not locally injective there, since z and $-z$ have the same image arbitrarily close to zero.

4.3 A logarithm is a function on a specified domain

A logarithm branch on a domain $\Omega \subset \mathbb{C} \setminus \{0\}$ is a holomorphic L with $\mathrm{e}^{L(z)} = z$. Necessarily

$$L(z) = \log|z| + \mathrm{i}\theta(z), \qquad L'(z) = 1/z.$$

The derivative follows from $\mathrm{e}^L L' = 1$. On a connected domain, two branches differ by a constant $2\pi\mathrm{i}k$: their difference is continuous and has values in the discrete set $2\pi\mathrm{i}\mathbb{Z}$.



Principal logarithm: $\mathbb{C} \setminus (-\infty, 0]$

Figure 4.1: A cut removes loops that force the argument to jump by 2π. The cut is not part of the logarithm's holomorphic domain.

The principal holomorphic logarithm is

$$\operatorname{Log} z = \log|z| + \mathrm{i}\operatorname{Arg} z, \qquad -\pi < \operatorname{Arg} z < \pi, \quad z \in \mathbb{C} \setminus (-\infty, 0].$$

On this domain $u_x = x/r^2$, $u_y = y/r^2$, $v_x = -y/r^2$, $v_y = x/r^2$, so C–R holds and $\operatorname{Log}' = 1/z$. Approaching a negative real point from above and below gives imaginary limits π and $-\pi$. A pointwise assignment of argument π on the ray does not remove this discontinuity.

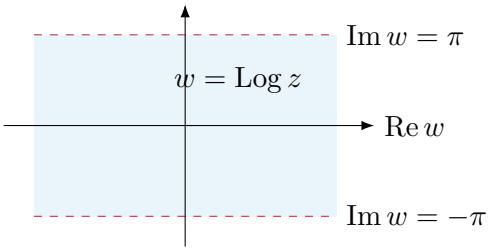


Figure 4.2: The principal logarithm maps the slit plane bijectively onto the open horizontal strip $-\pi < \operatorname{Im} w < \pi$.

Worked Example: The multiplication law needs a branch correction

Take $z = w = \mathrm{e}^{3\pi\mathrm{i}/4}$. Both are in the principal slit plane and $zw = -\mathrm{i}$ is also there. Yet $\operatorname{Log} z + \operatorname{Log} w = 3\pi\mathrm{i}/2$, whereas $\operatorname{Log}(zw) = -\pi\mathrm{i}/2$. Their difference is $2\pi\mathrm{i}$. The unconditional identity is $\mathrm{e}^{\operatorname{Log} z + \operatorname{Log} w} = zw$; equality of chosen logarithms needs the summed argument to stay in the chosen interval.

4.4 Global logarithms, primitives, and periods

No logarithm exists on $\mathbb{C} \setminus \{0\}$: integrating its supposed derivative on the unit circle would give both zero (a derivative around a closed path) and $\oint \mathrm{d}z/z = 2\pi\mathrm{i}$. More generally, a nonvanishing holomorphic g on a domain has a holomorphic logarithm if and only if

$$\oint_\gamma \frac{g'}{g}\mathrm{d}z = 0 \quad \text{for every closed contour } \gamma \subset \Omega.$$

Necessity is the chain rule. For sufficiency, the vanishing periods make $H(z) = \int_{z_*}^{z} g'/g$ path independent. Then $(g\mathrm{e}^{-H})' = 0$, so $g\mathrm{e}^{-H} = c \neq 0$. Choose one number ℓ with $\mathrm{e}^{\ell} = c$ and set $L = H + \ell$. Simple connectivity guarantees the period condition by Cauchy's theorem (Chapter 9).

4.5 Powers and inverse-function ranges

With a chosen logarithm, define $z^\alpha = \exp(\alpha L(z))$ and differentiate to get $\alpha z^\alpha/z$. Changing L by $2\pi\mathrm{i}k$ changes this value by $\mathrm{e}^{2\pi\mathrm{i}k\alpha}$. Integer powers are branch independent; general powers are not.

Worked Example: The principal square root and its exact image

On the principal slit plane, $s(z) = \exp(\operatorname{Log} z/2)$ has arguments in $(-\pi/2, \pi/2)$, so its image is the right half-plane, with no restriction on the imaginary coordinate. It is a bijection there, inverse $w \mapsto w^2$, and $s' = 1/(2s)$. On the upper half-plane alone, arguments lie in $(0, \pi)$, so the square root maps it to the first quadrant. These are different source domains and different images.

Worked Example: All values versus a chosen power

The multivalued expression i^{i} uses logarithms $\mathrm{i}(\pi/2 + 2\pi k)$ and gives positive real values $\exp[-\pi/2 - 2\pi k]$. The principal value is $\mathrm{e}^{-\pi/2}$. A question asking for all values is not answered by naming only the principal one.

Standard Workflow: Branch-sensitive work

Specify the source domain and argument interval before algebra. Find the image range, prove injectivity if an inverse is claimed, and differentiate within the branch domain. Track multiples of $2\pi\mathrm{i}$. To disprove a global logarithm, exhibit one nonzero period. To construct one, integrate g'/g only after justifying path independence.

4.6 Chapter Mind Map

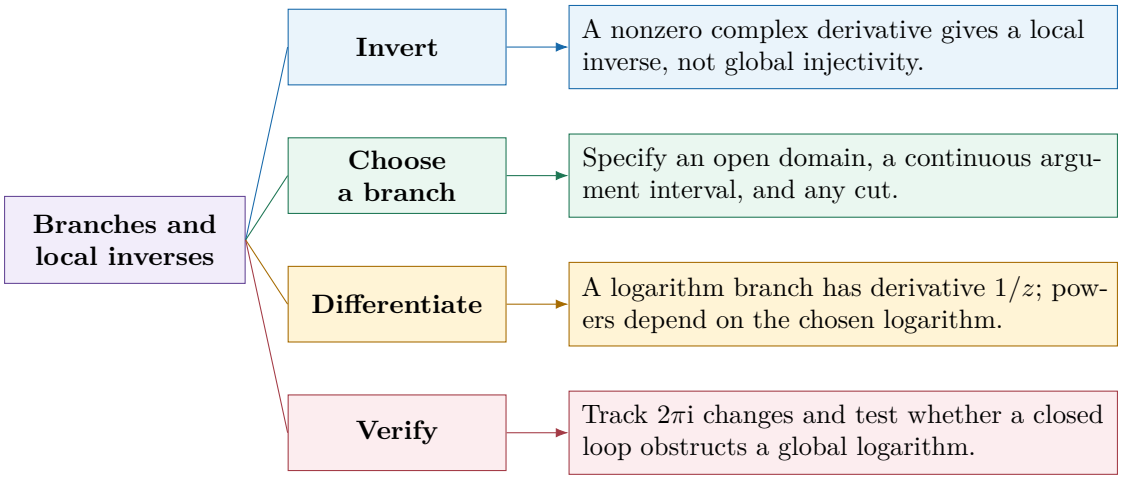


Figure 4.3: Concepts, calculation route, conditions, and checks.

5 Harmonic Functions and Conformal Geometry

Source map: Farinati, Chapter 5, pp. 22–29.

5.1 Why the Laplace equation appears

If $f = u + \mathrm{i}v$ is holomorphic and C^2, C–R gives

$$\Delta u = (v_y)_x + (-v_x)_y = 0, \qquad \Delta v = (-u_y)_x + (u_x)_y = 0.$$

The equality of mixed partials is the cancellation. Chapter 11 removes the extra C^2 qualification by proving holomorphic functions are smooth. Thus the real and imaginary parts provide two-dimensional Laplace solutions, connecting directly to the PDE T material.

5.2 Construct a harmonic conjugate without losing a function

For a prescribed harmonic u, seek v with $v_x = -u_y$, $v_y = u_x$. The one-form

$$\omega = -u_y \mathrm{d}x + u_x \mathrm{d}y$$

is closed because $\partial_x(u_x) - \partial_y(-u_y) = \Delta u = 0$. On a simply connected domain it is exact, so a global v exists. Locally the same is true on small disks. Two conjugates differ by a real constant on a connected domain.

Worked Example: A cubic harmonic part and complete integration

Let $u = x^3 - 3xy^2 + 2x$. Then $\Delta u = 6x - 6x = 0$. First integrate $v_x = -u_y = 6xy$ with respect to x:

$$v = 3x^2 y + C(y).$$

Next $v_y = 3x^2 + C'(y)$ must equal $u_x = 3x^2 - 3y^2 + 2$, so $C' = -3y^2 + 2$, $C = -y^3 + 2y + C_0$. Thus $v = 3x^2 y - y^3 + 2y + C_0$ and $f = z^3 + 2z + \mathrm{i}C_0$. Both C–R equations have now been checked, rather than only the equation used for integration.

Worked Example: A locally harmonic function with no global conjugate

$u = \log(x^2 + y^2) = 2\log|z|$ is harmonic on $\mathbb{C} \setminus \{0\}$. Its local conjugate is $v = 2\arg z$, not $\arg z$: the factor two matters. Along a positive circle,

$$\oint \omega = \oint \frac{-2y\mathrm{d}x + 2x\mathrm{d}y}{x^2 + y^2} = 4\pi \neq 0.$$

Therefore ω is not globally exact and there is no single-valued conjugate on the punctured plane. On a slit plane $f = 2\operatorname{Log} z$ works.

5.3 Conformal means a nonzero local complex multiplier

For curves γ_1, γ_2 through a with nonzero tangents v_1, v_2, their image tangents are $f'(a)v_1, f'(a)v_2$. If $f'(a) = \rho \mathrm{e}^{\mathrm{i}\theta} \neq 0$, both tangents rotate by θ and scale by ρ, so their oriented angle is preserved. Area is multiplied by $|f'(a)|^2$. At a critical point this argument fails; z^m multiplies small radial angles by m near zero rather than preserving them there.

Worked Example: Images of coordinate lines under a square

For $w = z^2 = u + \mathrm{i}v$, the line $x = 1$ maps to $u = 1 - y^2$, $v = 2y$, hence $u = 1 - v^2/4$. The line $y = 1$ maps to $u = x^2 - 1$, $v = 2x$, hence $u = v^2/4 - 1$. They meet at $z = 1 + \mathrm{i}$, where $f' = 2 + 2\mathrm{i} \neq 0$. Direct differentiation gives image tangents $(-2, 2)$ and $(2, 2)$, whose dot product is zero. The originally

Worked Example (continued)

perpendicular coordinate directions remain perpendicular.

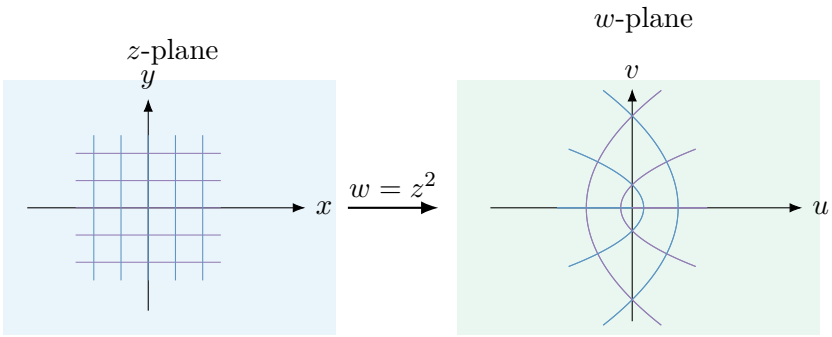


Figure 5.1: The square map sends a Cartesian grid to orthogonal parabola families away from its critical point at zero.

5.4 Complex potentials: keep the two conventions separate

For a holomorphic function $f = u + \mathrm{i}v$, the vector field $\mathbf{V} = (u, -v)$ has

$$\nabla \cdot \mathbf{V} = u_x - v_y = 0, \qquad \partial_x V_y - \partial_y V_x = -v_x - u_y = 0.$$

This is the source's incompressible, irrotational field convention. In potential-flow notation one instead starts with $W = \Phi + \mathrm{i}\Psi$ and sets the physical velocity to $\nabla\Phi = (\operatorname{Re} W', -\operatorname{Im} W')$. The potential is W, but its derivative is the complex velocity before conjugation. C–R implies $\nabla\Phi \cdot \nabla\Psi = 0$, so potential contours and streamlines are orthogonal at nonstagnation points.

Worked Example: A radial source and a global flux despite local zero divergence

Take $W = \operatorname{Log} z$ locally. Then $W' = 1/z$ and $\mathbf{V} = (x/r^2, y/r^2)$. It has zero divergence away from zero. On $r = R$, $\mathbf{V} \cdot \mathbf{n} = 1/R$ and $\mathrm{d}s = R\mathrm{d}\theta$, so the outward flux is 2π. This does not contradict Gauss: the disk contains the singularity, where the smooth divergence theorem hypotheses fail. An annulus has outer flux 2π and inner outward flux -2π, giving total zero.

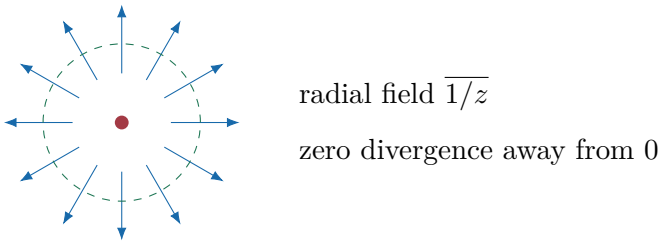


Figure 5.2: A singular source can carry nonzero net flux even though the divergence vanishes at every regular point.

5.5 Transfer a Laplace problem through a conformal map

If F is a real C^2 function on the image plane and $w = f(z)$ is holomorphic, the chain rule and C–R give

$$\Delta_z(F \circ f) = |f'(z)|^2 (\Delta_w F) \circ f.$$

The terms with first derivatives of F vanish because the components of f are harmonic; the mixed second-derivative terms cancel by orthogonality. Harmonicity is therefore preserved. To transfer a Dirichlet problem, map the domain, transform the boundary data by the inverse map, solve there, and compose back. Do not transform only the domain while leaving boundary data attached to the wrong points.

Worked Example: A sector problem solved by its angle

In $0 < \theta < \alpha$, prescribe $u = 0$ on the lower ray and $u = 1$ on the upper ray, and seek a bounded harmonic function. Then $u = \theta/\alpha$ is harmonic away from the vertex since $u_r = 0$, $u_{\theta\theta} = 0$. It is the imaginary part of $\operatorname{Log} z/\alpha$ on this sector. The boundary values at the vertex conflict, so no continuous value there is claimed. The mapping $z^{\pi/\alpha}$ turns the sector into a half-plane and gives the same angular formula.

Standard Workflow: Harmonic and geometric problems

Check $\Delta u = 0$; identify whether a global conjugate is justified; integrate one C–R equation with a one-variable integration function; impose the other; normalize with given data. For conformality test $f' \neq 0$, calculate image curves, and keep local and global injectivity distinct. For a physical field state whether f is a potential or a complex velocity.

5.6 Chapter Mind Map

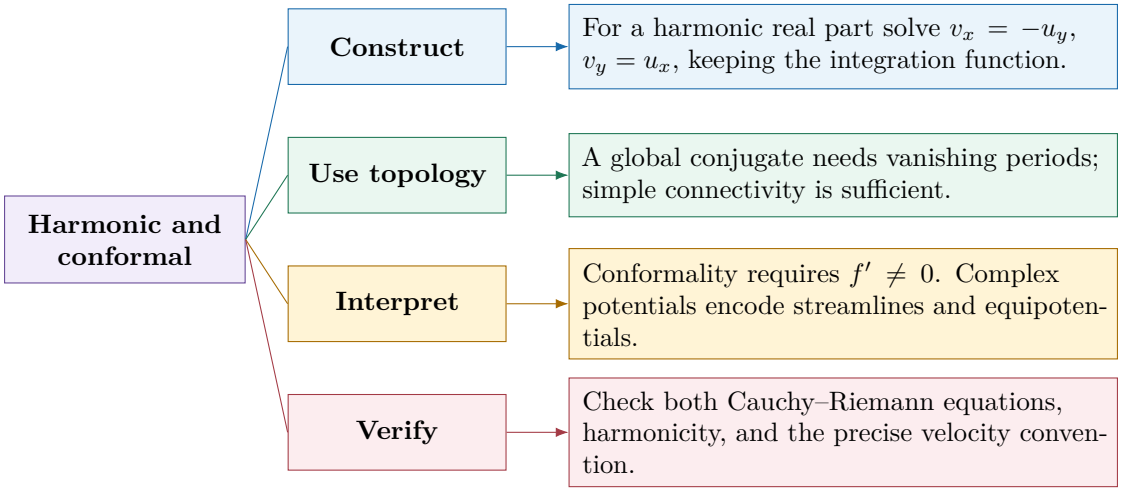


Figure 5.3: Concepts, calculation route, conditions, and checks.

6 Möbius Transformations and the Riemann Sphere

Source map: Farinati, Chapter 6, pp. 30–43.

6.1 One extra point makes inversion a bijection

The extended plane is $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. A neighborhood of infinity contains all sufficiently large $|z|$. The coordinate $w = 1/z$ turns a neighborhood of infinity into one of zero. Stereographic projection identifies this space with a sphere: the north pole represents infinity, while ordinary lines become circles through the north pole.

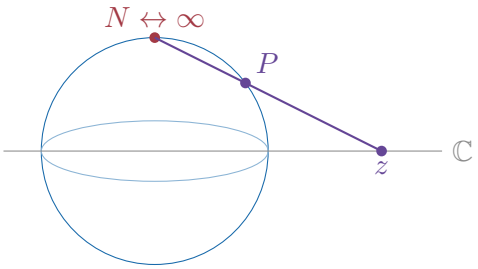


Figure 6.1: Stereographic projection explains why lines and circles form one geometric family on the extended plane.

The map $z \mapsto 1/z$ extends by $0 \mapsto \infty$, $\infty \mapsto 0$. Infinity is not a real number to be inserted into algebraic expressions; values there mean limits in the spherical coordinate. For the unit sphere, one explicit stereographic convention is

$$S(x+\mathrm{i}y) = \left(\frac{2x}{1+|z|^2}, \frac{2y}{1+|z|^2}, \frac{|z|^2-1}{1+|z|^2} \right), \qquad z = \frac{X+\mathrm{i}Y}{1-Z}.$$

The sum of the squared sphere coordinates is one, and $|z| \to \infty$ sends $S(z)$ to the north pole $(0,0,1)$. The chordal distance convention

$$\chi(z,w) = \frac{|z-w|}{\sqrt{1+|z|^2}\sqrt{1+|w|^2}}, \qquad \chi(z,\infty) = \frac{1}{\sqrt{1+|z|^2}}$$

is half the Euclidean distance of the sphere images. In particular $z_n \to \infty$ on the sphere if and only if $|z_n| \to \infty$, regardless of direction. This compactifies all unbounded directions by one point, not by separate positive and negative infinities.

6.2 Matrix composition and all exceptional values

Definition: Möbius transformation

For $ad - bc \neq 0$, set $T(z) = (az+b)/(cz+d)$. Its representing matrix is $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, up to multiplication by a nonzero scalar.

If $c \neq 0$, the pole is $-d/c$, $T(-d/c) = \infty$, and $T(\infty) = a/c$. If $c = 0$, then $a, d \neq 0$, the map is affine, and $T(\infty) = \infty$. Its inverse is

$$T^{-1}(w) = \frac{dw-b}{-cw+a}, \qquad T'(z) = \frac{ad-bc}{(cz+d)^2}.$$

Direct substitution proves the inverse; direct expansion proves $T_A \circ T_B = T_{AB}$. Thus composition is matrix multiplication, but two scalar-multiple matrices give the same map: this is projective matrix algebra, not a unique matrix representation.

Worked Example: Composition without expanding nested fractions

For $S(z) = z+2$ and $T(z) = 1/z$, the matrices are $\left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Hence $S \circ T = (2z+1)/z$, while $T \circ S = 1/(z+2)$. The different products show that composition order matters.

6.3 Why generalized circles stay generalized circles

A generalized circle has an equation

$$A|z|^2 + Bz + \bar{B}\bar{z} + C = 0, \qquad A, C \in \mathbb{R}.$$

When $A = 0$ and $B \neq 0$, this is a line; for $A \neq 0$ it is a circle when its completed-square radius is positive. Substitute $z = 1/w$ and multiply by $|w|^2$ to obtain another equation of the same kind. Translations and

nonzero complex scalings also preserve this form. For $c \neq 0$,

$$T(z) = \frac{a}{c} + \frac{bc - ad}{c^2} \frac{1}{z + d/c},$$

so every Möbius map is a composition of translations, scalings, and inversion. This proves circle preservation without assuming that every circle can be moved by scaling alone to one special case.

Worked Example: Inversion changes a line to a circle

For $z = 1 + it$, $w = 1/z = (1 - it)/(1 + t^2)$. Then $|w - 1/2| = 1/2$. Thus the finite line maps to that circle with $w = 0$ omitted; including its point at infinity fills in zero. A finite-image statement must not accidentally add the omitted point.

6.4 Three-point construction and the side test

For three distinct finite points, the map

$$Q(z) = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}$$

sends (z_1, z_2, z_3) to $(0, 1, \infty)$. If a target triple is prescribed, form its corresponding P and use $P^{-1} \circ Q$. Infinity entries are handled by limits. Uniqueness follows because a Möbius map fixing $0, 1, \infty$ is the identity.

To map a boundary, three distinct points determine the image generalized circle. To map a *region*, test one interior point or derive a modulus inequality. Boundary data alone do not tell you which side is the image.

Worked Example: The upper half-plane to the disk

Let $C(z) = (z - \mathrm{i})/(z + \mathrm{i})$. For real x, $|C(x)| = 1$, and $C(\mathrm{i}) = 0$. More explicitly,

$$|z + \mathrm{i}|^2 - |z - \mathrm{i}|^2 = 4 \operatorname{Im} z,$$

so $\operatorname{Im} z > 0$ is equivalent to $|C(z)| < 1$. The inverse is $z = \mathrm{i}(1 + w)/(1 - w)$, whose imaginary part is $(1 - |w|^2)/|1 - w|^2 > 0$. This proves onto-ness, not only containment. The boundary point at infinity maps to 1.

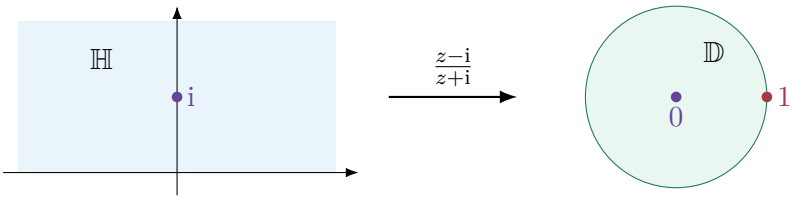


Figure 6.2: A single interior point selects the disk rather than its exterior; the inverse inequality verifies the whole image.

6.5 An entire region, not only its four boundary curves

Worked Example: The image of an open square

Let $T(z) = (z + 1)/(z - 1)$ and $S = \{0 < x < 1,\ 0 < y < 1\}$. Write $w = u + \mathrm{i}v$. Since $T^{-1} = T$,

$$x = \frac{u^2 + v^2 - 1}{D}, \qquad y = \frac{-2v}{D}, \qquad D = (u - 1)^2 + v^2 > 0.$$

Worked Example (continued)

The four strict inequalities become

$$|w| > 1, \quad u < 1, \quad v < 0, \quad (u-1)^2 + (v+1)^2 > 1.$$

These conditions are equivalent to membership in $T(S)$, because the displayed inverse reconstructs a point in the square. The boundaries are the real line, the unit circle, the line $u = 1$, and the circle centered at $1 - \mathrm{i}$ of radius one, with the correct arcs selected by the other inequalities. The corner $z = 1$ is not in the open square but causes an unbounded image near it.

Standard Workflow: Möbius questions

Check the determinant first; list the pole and infinity image; construct or compose using matrices; map three boundary points; determine the side using an interior point; for a polygon derive inverse inequalities to select the correct arcs. Verify every prescribed image and state whether the extended or finite plane is intended.

6.6 Chapter Mind Map

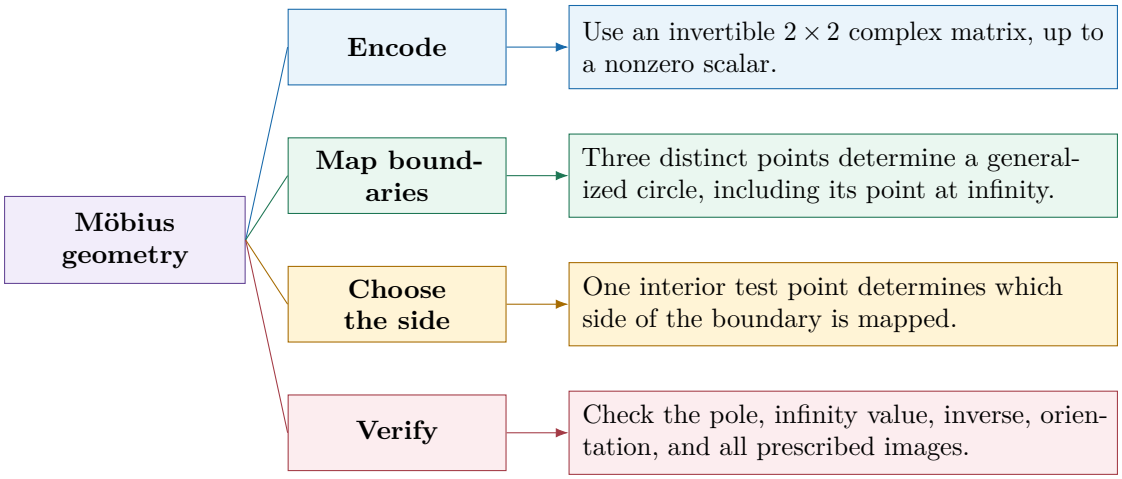


Figure 6.3: Concepts, calculation route, conditions, and checks.

7 Complex Series, Radii, and Boundary Convergence

Source map: Farinati, Chapter 7, pp. 44–60.

7.1 Absolute convergence controls tails

A complex series converges when its partial sums converge. If $\sum |a_n| < \infty$, then

$$\left| \sum_{n=p}^{q} a_n \right| \leq \sum_{n=p}^{q} |a_n| \to 0 \quad (p \to \infty),$$

uniformly in $q \geq p$. Completeness proves convergence. The necessary condition $a_n \to 0$ alone is not sufficient: $\sum 1/n$ diverges.

7.2 Root and ratio tests, including missing coefficients

Let $L = \limsup |a_n|^{1/n}$. If $L < 1$, choose $L < r < 1$; eventually $|a_n| \le r^n$, proving absolute convergence. If $L > 1$, choose $1 < r < L$; infinitely many terms have $|a_n| > r^n$, so they do not tend to zero. This does not claim every term tends to infinity when a limsup exceeds one. At $L = 1$, the test is inconclusive.

If the nonzero-term ratios $|a_{n+1}/a_n|$ are defined eventually and have limit L, repeated bounds by $L \pm \varepsilon$ show that the root limit is also L (with one-sided bounds when $L = 0$). A ratio formula cannot be used through infinitely many zero coefficients.

Key Result: Cauchy–Hadamard radius

For $\sum a_n(z-a)^n$,

$$R = \frac{1}{\limsup |a_n|^{1/n}}, \qquad 1/0 = \infty, \quad 1/\infty = 0.$$

The series converges absolutely for $|z-a| < R$ and diverges for $|z-a| > R$. Every point on the boundary requires separate analysis.

This follows by applying the root test to $a_n(z-a)^n$. If the series converges at some noncentral z_0, its terms are bounded, so for $|z-a| < |z_0 - a|$ they are bounded by a constant times a geometric sequence. This is another direct explanation of the disk shape.

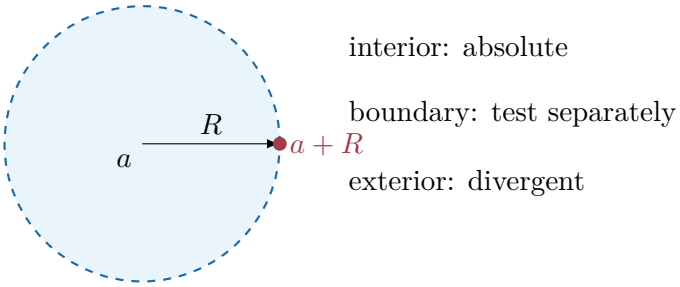


Figure 7.1: A radius does not classify its boundary. A picture of a convergence set must mark boundary exceptions.

Worked Example: Factorials and a ratio limit

For $a_n = (2n)!/(n!)^2$, $a_{n+1}/a_n = (2n+2)(2n+1)/(n+1)^2 \to 4$, so $\sum a_n z^n$ has $R = 1/4$. For $\sum n! z^n$, the term ratio is $(n+1)|z|$ and $R = 0$. For $\sum z^n/n!$, it is $|z|/(n+1)$ and $R = \infty$. The point $z = 0$ must still be included in the radius-zero case.

Worked Example: A sparse series where the ratio test is unusable

$\sum_{k \ge 0} z^{2k}$ has coefficients $a_{2k} = 1$, $a_{2k+1} = 0$. The ordinary coefficient-root sequence oscillates, but its limsup is one, so $R = 1$. Equivalently it is a geometric series in z^2. On $|z| = 1$, the terms have modulus one and never tend to zero, so every boundary point diverges.

7.3 Summation by parts explains conditional convergence

Suppose b_n has partial sums bounded by M, and $a_n \downarrow 0$ are nonnegative real numbers. For a tail let $A_n = \sum_{k=p}^{n} b_k$, so $|A_n| \le 2M$. Summation by parts gives

$$\sum_{n=p}^{q} a_n b_n = a_q A_q + \sum_{n=p}^{q-1} (a_n - a_{n+1}) A_n.$$

The modulus is at most $2M[a_q + a_p - a_q] = 2Ma_p \to 0$. This proves Dirichlet's criterion. On $|z| = 1$, $z \neq 1$,

$$\left|\sum_{n=0}^{N} z^n\right| = \left|\frac{1 - z^{N+1}}{1 - z}\right| \leq \frac{2}{|1 - z|}.$$

Thus oscillation supplies cancellation that absolute-value estimates would discard.

Worked Example: The whole boundary of a family

For $p > 0$, consider $\sum_{n\geq 1} z^n/n^p$. It has $R = 1$. On the circle, absolute convergence holds exactly when $p > 1$. If $0 < p \leq 1$, Dirichlet gives conditional convergence at every $z \neq 1$, while $z = 1$ is the divergent p-series. For $p \leq 0$, the terms on the circle fail to tend to zero. These are separate cases; writing only $R = 1$ misses most of the question.

7.4 Composite series give geometric regions

Worked Example: A convergence disk exterior after substitution

For $\sum_{n\geq 0}[(3z + 1)/(6z - 2)]^n$, convergence is equivalent to $|3z + 1| < |6z - 2|$, with $z \neq 1/3$. Write $z = x + \mathrm{i}y$ and expand:

$$27(x^2 + y^2) - 30x + 3 > 0 \iff (x - 5/9)^2 + y^2 > 16/81.$$

The convergence region is outside the closed circle centered at $5/9$ of radius $4/9$. The undefined point $1/3$ is already inside the excluded disk. Every boundary point gives a geometric ratio of modulus one and diverges.

Worked Example: Absolute convergence of a complex-exponent series

For $\sum_{n\geq 1} n^{-s}$, define $n^{-s} = \exp(-s \log n)$ with the real $\log n$. Then $|n^{-s}| = n^{-\operatorname{Re} s}$, so absolute convergence holds exactly when $\operatorname{Re} s > 1$. This is a half-plane in the parameter s, not a power-series disk. No analytic continuation is needed for this assertion.

Standard Workflow: Convergence-set questions

Identify the true summation variable and center; compute a valid root limsup or ratio; separately test every type of boundary point; record absolute versus conditional convergence; for a composite expression pull the convergence set back by an exact inequality; exclude undefined expressions before drawing the final region.

7.5 Chapter Mind Map

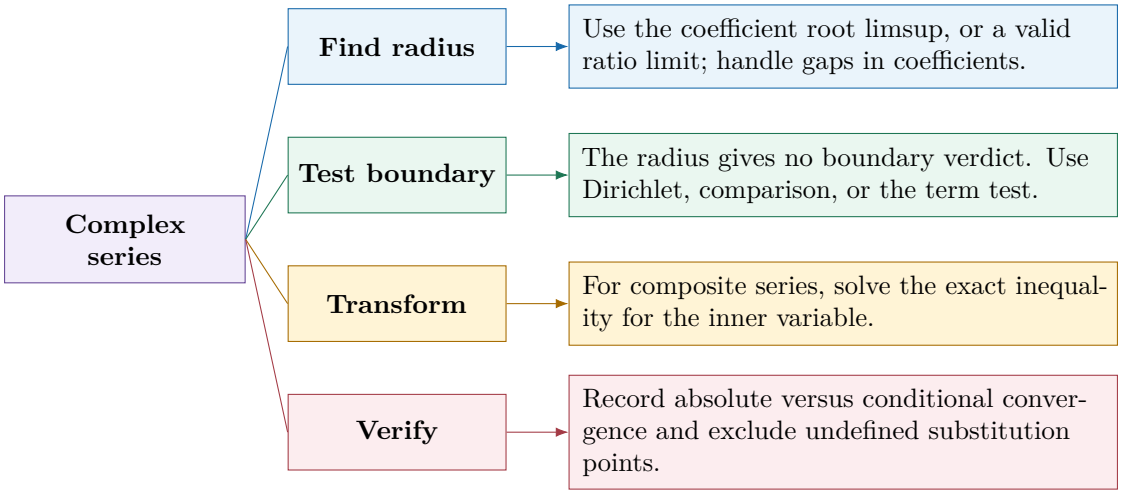


Figure 7.2: Concepts, calculation route, conditions, and checks.

8 Function Sequences and Holomorphic Limits

Source map: Farinati, Chapter 8, pp. 61–64; 9.7 and 10.1, pp. 75, 80–84.

8.1 The order of quantifiers is the distinction

Pointwise convergence means that for each fixed z and $\varepsilon > 0$, some $N(z, \varepsilon)$ controls the tail. Uniform convergence on E means one $N(\varepsilon)$ works for every $z \in E$, equivalently

$$\sup_{z \in E} |f_n(z) - f(z)| \to 0.$$

Local uniform convergence means uniform convergence on every compact subset of an open domain. It is weaker than uniform convergence on the whole domain but strong enough for holomorphic limit theorems.

Worked Example: The same sequence on three different sets

$f_n(z) = z^n$ tends to zero on $\mathbb{D}$. On $|z| \le r < 1$, $\sup |f_n| = r^n \to 0$, so convergence is locally uniform. On all of $\mathbb{D}$, the supremum is one for every n, so it is not uniform. On $[0, 1]$, the pointwise limit is zero below one and one at one, hence discontinuous; uniform convergence of continuous functions would have preserved continuity.

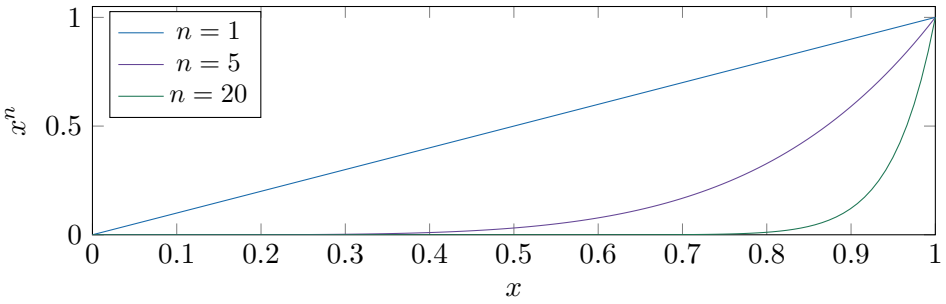


Figure 8.1: Boundary layers in x^n show why a point-dependent convergence speed is not a uniform bound.

8.2 The M-test and explicit error control

If $|g_n(z)| \le M_n$ on E and $\sum M_n < \infty$, then

$$\sup_E \left| \sum_{n=p}^{q} g_n \right| \le \sum_{n=p}^{q} M_n \to 0.$$

Thus the series converges uniformly and absolutely. This proves the M-test directly by uniform Cauchy tails. For a power series with radius R, choose $r < \rho < R$; boundedness of $a_n \rho^n$ gives a geometric majorant on $|z-a| \le r$. The derivative series has the same radius because $n^{1/n} \to 1$; repeated derivatives also converge locally uniformly inside the disk.

Worked Example: A useful numerical truncation bound

For $\sum z^n$ on $|z| \le r < 1$, truncation after degree N has error at most $r^{N+1}/(1-r)$. On $r = 1/2$, an error below 10^{-3} is guaranteed by $2^{-N} < 10^{-3}$, so $N = 10$ suffices. A convergence theorem becomes useful for computation when its tail estimate is retained.

8.3 When a limit may pass through an integral

For a fixed rectifiable contour γ of length L, uniform convergence on its image gives

$$\left| \int_\gamma f_n \mathrm{d}z - \int_\gamma f \mathrm{d}z \right| \le L \sup_\gamma |f_n - f| \to 0.$$

Uniformity on the whole domain is unnecessary; uniformity on the contour is enough. Multiplication by a bounded factor preserves this convergence. In particular, for $a \notin \gamma$, $1/(z-a)^k$ is bounded on the compact curve, so weighted contour integrals also pass to the limit.

Worked Example: Termwise integration with a genuine bound

On $|z| = r < 1$, the geometric series is uniformly convergent, so

$$\oint_{|z|=r} \frac{\mathrm{d}z}{z(1-z)} = \sum_{n \ge 0} \oint_{|z|=r} z^{n-1} \mathrm{d}z = 2\pi \mathrm{i}.$$

The multiplier $1/z$ is bounded on this fixed circle. This argument cannot be justified by merely saying the series converges pointwise at every point of the curve.

8.4 Why local uniform holomorphic limits stay holomorphic

Let f_n be holomorphic on Ω and converge locally uniformly to f. Fix a closed disk $\overline{D(a,R)} \subset \Omega$. Uniform convergence gives continuity of f and lets the Cauchy formula pass to the limit:

$$f(z) = \frac{1}{2\pi \mathrm{i}} \oint_{|w-a|=R} \frac{f(w)}{w-z} \mathrm{d}w \qquad (|z-a| < R).$$

The kernel difference quotient converges uniformly on the fixed circle when z varies in a smaller disk, because the denominator has a positive distance from zero. Therefore differentiating the integral gives a complex derivative and, repeatedly,

$$f^{(k)}(z) = \frac{k!}{2\pi \mathrm{i}} \oint \frac{f(w)}{(w-z)^{k+1}} \mathrm{d}w.$$

The same estimates show $f_n^{(k)} \to f^{(k)}$ uniformly on every compact subset. For a general compact set use finitely many such smaller disks. This proof uses Cauchy's formula, established in Chapter 10, rather than assuming that uniform convergence preserves derivatives in all settings.

Worked Example: The false real-analysis analogy

On $\mathbb{R}$, $g_n(x) = \sin(nx)/n$ converges uniformly to zero since $|g_n| \le 1/n$. Yet $g_n'(0) = 1$, whereas the limit derivative is zero. These functions are entire when extended to z, but they do not converge locally uniformly on complex neighborhoods: $|\sin(niy)|/n = \sinh(n|y|)/n$ grows for fixed $y \neq 0$. There is no conflict with the holomorphic theorem.

8.5 Morera's converse and normal-family vocabulary

If a continuous f has zero integral around every triangle compactly contained in an open set, then it is holomorphic. On a small disk, define $F(z)$ by integrating from a fixed point along a segment. Triangle cancellation makes the difference $F(z+h) - F(z)$ the integral on the short segment from z to $z+h$. Dividing by h and using continuity yields $F' = f$. Cauchy's formula then makes F' holomorphic. This proves Morera's theorem and offers another proof of the holomorphic limit theorem by passing triangle integrals to the limit.

A family is *normal* if every sequence has a subsequence converging locally uniformly (in the finite-valued sense used here). Montel's theorem says locally uniformly bounded holomorphic families are normal. Chapter 15 explains the compactness mechanism and its role in Riemann mapping; arbitrary uniformly bounded families of discontinuous functions do not enjoy this conclusion.

Standard Workflow: Limit interchange

State the set on which convergence is asserted. Produce a supremum bound or a summable majorant. Use contour length for integrals. For derivatives, invoke holomorphy and local uniform convergence on an open neighborhood, not uniformity on a real segment. Track the smaller compact disk needed for a positive denominator distance.

8.6 Chapter Mind Map

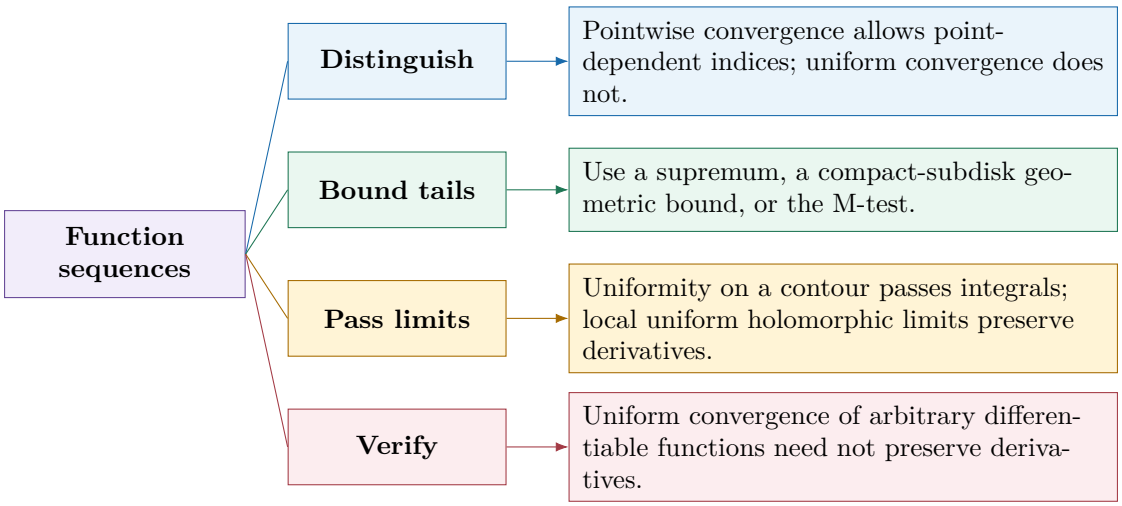


Figure 8.2: Concepts, calculation route, conditions, and checks.

9 Contour Integrals, Primitives, and Cauchy's Theorem

Source map: Farinati, Chapter 9, pp. 65–76; Goursat's proof added to make the foundational implication explicit.

9.1 Definition and orientation

For a piecewise C^1 parametrization $\gamma : [a, b] \to \mathbb{C}$, define

$$\int_\gamma f(z)\mathrm{d}z = \int_a^b f(\gamma(t))\gamma'(t)\mathrm{d}t.$$

An orientation-preserving change of parameter gives the same result by substitution; reversing orientation changes the sign. If $|f| \le M$ on the path, the triangle inequality gives the ML bound

$$\left|\int_\gamma f\mathrm{d}z\right| \le \int_a^b |f(\gamma)||\gamma'|\mathrm{d}t \le M \text{ length}(\gamma).$$

Worked Example: A nonholomorphic integrand can still be integrated

On $\gamma(t) = R\mathrm{e}^{\mathrm{i}t}$, $0 \le t \le 2\pi$, $\bar{z} = R\mathrm{e}^{-\mathrm{i}t}$ and $\mathrm{d}z = \mathrm{i}R\mathrm{e}^{\mathrm{i}t}\mathrm{d}t$. Hence $\oint \bar{z}\mathrm{d}z = 2\pi\mathrm{i}R^2$. It is not zero because $\bar{z}$ is not holomorphic. On a straight segment from zero to a, $z = ta$ instead gives $\int \bar{z}\mathrm{d}z = |a|^2/2$.

9.2 A complex integral packages two real integrals

Writing $f = u + \mathrm{i}v$ and $\mathrm{d}z = \mathrm{d}x + \mathrm{i}\mathrm{d}y$ gives

$$\int f\mathrm{d}z = \int (u\mathrm{d}x - v\mathrm{d}y) + \mathrm{i}\int (v\mathrm{d}x + u\mathrm{d}y).$$

For $\mathbf{V} = (u, -v)$, these are circulation and the right-normal flux. On a positive simple closed contour, the right normal is outward. Green's theorem makes both zero when f is C^1 holomorphic throughout the enclosed region. This is a useful link to Calculus 2T, but a hole or singularity invalidates the filled-region argument.

9.3 Primitives are equivalent to vanishing periods

If $F' = f$, the chain rule and the real fundamental theorem give $\int_\gamma f = F(\gamma(b)) - F(\gamma(a))$. Thus all closed-path integrals vanish. Conversely, if every closed-path integral vanishes on a domain, define $F(z) = \int_a^z f$; differences of paths are closed, so this is independent of path. On a small final straight segment,

$$\frac{F(z+h) - F(z)}{h} = \int_0^1 f(z + th)\mathrm{d}t \to f(z),$$

by continuity. This constructs the primitive. Path independence, zero periods, and existence of a primitive are therefore equivalent.

Worked Example: The obstruction carried by one power

For integer $n \ne -1$, $z^{n+1}/(n+1)$ is a primitive of z^n on its natural domain. For $n = -1$, a circle $z = a + R\mathrm{e}^{\mathrm{i}t}$ gives

$$\oint \frac{\mathrm{d}z}{z-a} = \int_0^{2\pi} \mathrm{i}\mathrm{d}t = 2\pi\mathrm{i}.$$

Thus $1/(z-a)$ has no global primitive on a punctured plane. A local branch of $\mathrm{Log}(z-a)$ is not a global solution to that problem.

9.4 Goursat's triangle argument without continuous derivatives

Let f be holomorphic on a neighborhood of a filled triangle T, and set $I(T) = \int_{\partial T} f$. Divide T into four similar triangles; internal edges cancel. At least one child has $|I(T_1)| \geq |I(T)|/4$. Repeat, obtaining nested closed triangles T_n with diameters $d/2^n$ and perimeters $L/2^n$, and $|I(T_n)| \geq |I(T)|/4^n$. Their common point is a.

Differentiability at a gives $f(z) = f(a) + f'(a)(z-a) + (z-a)\varepsilon(z)$ with $\varepsilon(z) \to 0$. The constant and linear terms have primitives, so their integrals around T_n vanish. For large n, $|\varepsilon| \leq \epsilon$ on T_n, hence

$$|I(T_n)| \leq \epsilon \frac{d}{2^n} \frac{L}{2^n}.$$

Combining gives $|I(T)| \leq \epsilon dL$. Since ϵ is arbitrary, $I(T) = 0$. No continuity of f' was used.

Triangle cancellation and subdivision extend this result to null-homotopic contours. In particular, every holomorphic function on a simply connected domain has zero closed-path integrals and a primitive. A standard equivalent statement is that an integral is unchanged under a contour homotopy staying within the holomorphy domain. Piecewise smooth contours can be replaced by nearby polygonal contours in local disks; their short connecting pieces cancel. The topological condition concerns the entire deformation, not just its endpoints.

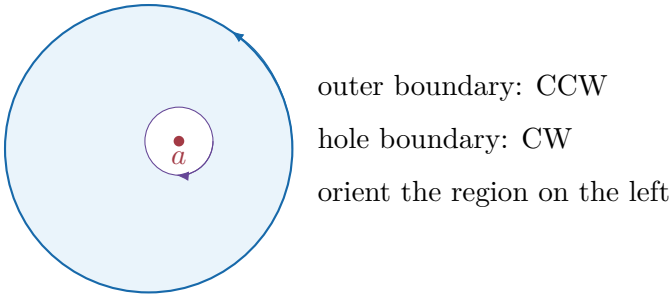


Figure 9.1: Removing a small disk creates an inner clockwise boundary. Moving it to the other side of the equality yields a positive small-circle integral.

9.5 Winding number records the missing topology

For a closed contour avoiding a,

$$\mathrm{Ind}(\gamma, a) = \frac{1}{2\pi \mathrm{i}} \int_\gamma \frac{\mathrm{d}z}{z-a}.$$

Choose a continuous argument of $\gamma(t) - a$ along the parameter interval (not necessarily on the plane). Its net change is $2\pi k$, because the endpoints coincide. The change in $\log|\gamma - a|$ is zero, so integrating the logarithmic differential proves $\mathrm{Ind} = k \in \mathbb{Z}$. It is constant on each component of the complement of the curve; it is zero sufficiently far away. A simple positive circle has index one inside and zero outside; repeated traversal multiplies it.

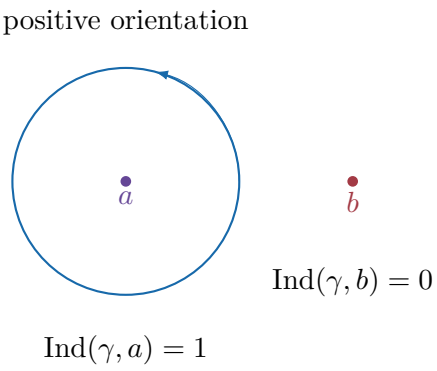


Figure 9.2: "Inside" generalizes to an integer winding number when curves are not simple.

Worked Example: Deformation cannot cross a pole

For a positive circle of radius two centered at zero, $\oint \mathrm{d}z/(z-1) = 2\pi\mathrm{i}$. For radius $1/2$ the value is zero. One cannot shrink the first circle to the second through curves avoiding 1: at some stage the deformation must cross the pole. At radius one the contour passes through the pole, and the ordinary integral is undefined.

Standard Workflow: Choose the cheapest valid contour method

Parameterize directly if conjugates or $|z|$ occur on a simple path. Use a primitive for endpoint data. Use Cauchy's theorem only after checking the filled region or null-homotopy. Use winding for $1/(z-a)$. Record orientation and never deform across a singularity. An ML estimate is also a proof, not merely an approximation slogan.

9.6 Chapter Mind Map

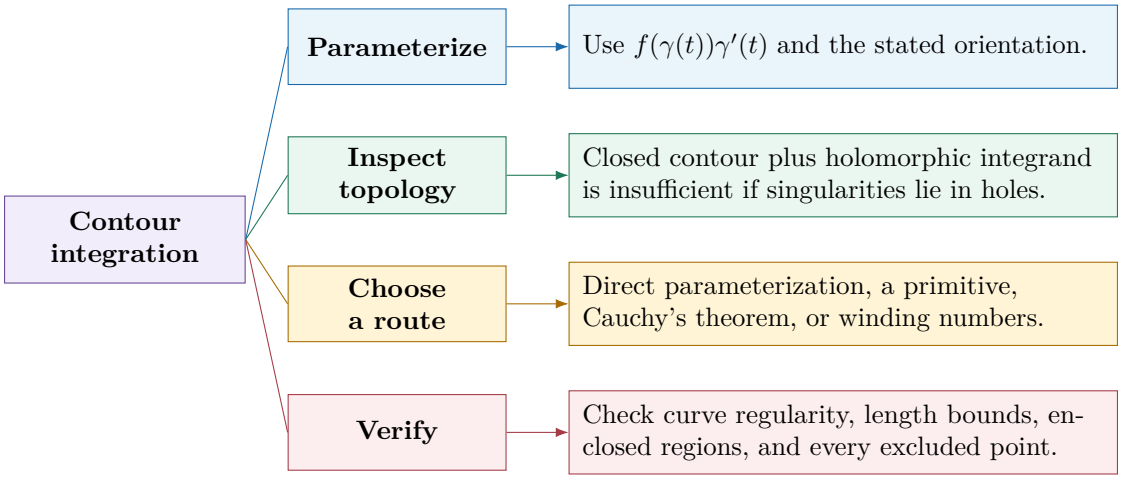


Figure 9.3: Concepts, calculation route, conditions, and checks.

10 Cauchy Formulas, Estimates, and Rigidity

Source map: Farinati, Chapter 10, pp. 77–90; higher derivatives in Chapter 11.

10.1 Boundary values determine the interior

Let f be holomorphic on a neighborhood of a closed Jordan region bounded positively by γ, and let a be inside. Remove a small circle about a and use Cauchy's theorem on the intervening region. Then

$$\oint_{\gamma} \frac{f(z)}{z-a}\mathrm{d}z = \oint_{|z-a|=r} \frac{f(z)}{z-a}\mathrm{d}z.$$

Subtract $f(a)\oint \mathrm{d}z/(z-a) = 2\pi\mathrm{i}f(a)$. The remainder is bounded by

$$\frac{\sup_{|z-a|=r}|f(z)-f(a)|}{r}\,2\pi r \to 0.$$

The left side is independent of r, so it equals its limiting value.

Key Result: Cauchy's integral formula and its derivatives

$$f(a)=\frac{1}{2\pi\mathrm{i}}\oint_\gamma \frac{f(z)}{z-a}\mathrm{d}z, \qquad f^{(n)}(a)=\frac{n!}{2\pi\mathrm{i}}\oint_\gamma \frac{f(z)}{(z-a)^{n+1}}\mathrm{d}z.$$

The numerator must be holomorphic on the relevant filled region. A boundary point is not an admissible a in this formula. For general admissible cycles, winding numbers replace the simple inside/outside factor.

To differentiate the first formula, keep the contour fixed and restrict a to a smaller interior disk. The difference quotient of $1/(z-a)$ converges uniformly there to $1/(z-a)^2$, since denominators stay bounded away from zero. Repeating gives the factorial $n!$. This both proves the derivative formula and justifies automatic higher differentiability.

Worked Example: Location, order, and factorial in one calculation

For positive $|z|=2$,

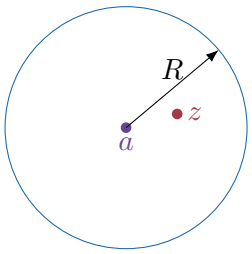
$$\oint \frac{\mathrm{e}^{2z}}{(z-\mathrm{i})^4}\mathrm{d}z=\frac{2\pi\mathrm{i}}{3!}(\mathrm{e}^{2z})^{(3)}\big|_{z=\mathrm{i}}=\frac{8\pi\mathrm{i}}{3}\mathrm{e}^{2\mathrm{i}}.$$

The point i is inside and the remaining numerator is entire. Reversing orientation negates the result. On $|z|=1$, the denominator vanishes on the contour, so no ordinary integral is defined.

10.2 Cauchy estimates and three rigidity applications

On $|z-a|=R$, if $|f|\le M_R$, the derivative formula and ML give

$$|f^{(n)}(a)|\le \frac{n!M_R}{R^n}.$$



$|f^{(n)}(a)|\le n!M_R/R^n$

the circle stays in the domain

Figure 10.1: A contour formula becomes a derivative estimate by dividing by a fixed power of the radius.

Liouville. If f is entire and globally bounded by M, then $|f'(a)|\le M/R$ for every R. Let $R\to\infty$; $f'=0$, and connectedness makes f constant.

Fundamental theorem of algebra. Suppose a nonconstant polynomial p had no zero. Then $1/p$ would be entire. Outside a large disk, the leading term gives $|p(z)|\ge |a_n||z|^n/2$, so $1/p$ is bounded there. Inside the disk it is bounded by continuity and compactness. Liouville makes $1/p$, hence p, constant, a contradiction. Repeated polynomial division then gives all roots with multiplicity, linking analysis back to Algebra 2M.

Polynomial growth. If f is entire and $|f(z)|\le C(1+|z|)^m$ for integer $m\ge 0$, then for each fixed a,

$$|f^{(m+1)}(a)|\le \frac{(m+1)!C(1+|a|+R)^m}{R^{m+1}}\to 0.$$

Therefore f is a polynomial of degree at most m. The bound is global in all complex directions, not merely along the real axis.

Worked Example: A one-sided real-part bound is enough

If f is entire and $\operatorname{Re} f \le A$, then $|e^f| \le e^A$. Liouville makes e^f constant. Differentiating gives $e^f f' = 0$; since the exponential never vanishes, $f' = 0$. A lower real-part bound is handled with e^{-f}. Merely being bounded on the real axis would not suffice: $\sin z$ is a counterexample.

Worked Example: Strictly sublinear growth

If f is entire and $|f(z)|/|z| \to 0$ as $|z| \to \infty$, fix a and $\epsilon > 0$. Outside a fixed disk, $|f(z)| \le \epsilon|z|$, while on that disk it has some bound M_ϵ. Hence $M_R \le M_\epsilon + \epsilon(|a| + R)$ on the radius-R circle about a. Cauchy's estimate gives $|f'(a)| \le M_\epsilon/R + \epsilon(1 + |a|/R)$. First let $R \to \infty$, then $\epsilon \to 0$. Thus f is constant. Taking these limits in the wrong order would hide the dependence of M_ϵ.

10.3 Maximum and minimum modulus principles

If $|f|$ has a local maximum M at a, apply the circle form $f(a) = (2\pi)^{-1} \int f(a + re^{it})dt$ in a small disk. When $M > 0$, rotate the values so $f(a) = M$ is real. Each real part on the circle is at most M, but its average is M; continuity forces every real part to equal M. The modulus bound then forces every imaginary part to vanish. Varying r makes f constant on the disk. If $M = 0$, it is already zero there. The set of points attaining the maximal modulus is open and closed within a connected region where that bound holds, giving the strong principle; equivalently use the identity theorem proved next.

Thus a nonconstant holomorphic function has no interior local modulus maximum. On a bounded domain, continuity up to the compact boundary gives a maximum there. If f never vanishes, apply the same theorem to $1/f$ to get the minimum modulus principle. Zeros are genuine exceptions to the latter.

Worked Example: A boundary quotient bound and an equality case

Suppose f, g are holomorphic near $\overline{\mathbb{D}}$, g has no zeros on that closed disk, and $|f| \le 5|g|$ on its boundary. Then $h = f/g$ is holomorphic and $|h| \le 5$ inside. If $f(a) = 5g(a)$ at one interior point, h attains its maximum modulus, so $h \equiv 5$ and $f = 5g$. Without the nonvanishing hypothesis on g, the quotient argument may fail.

Standard Workflow: Cauchy and entire-function problems

Mark every denominator zero and locate it relative to the contour; factor out a holomorphic numerator; match power $n + 1$ to derivative n and divide by $n!$. For rigidity choose a circle with variable radius, prove the required global bound, and send the radius to infinity. State connectedness, nonvanishing, and equality cases explicitly.

10.4 Chapter Mind Map

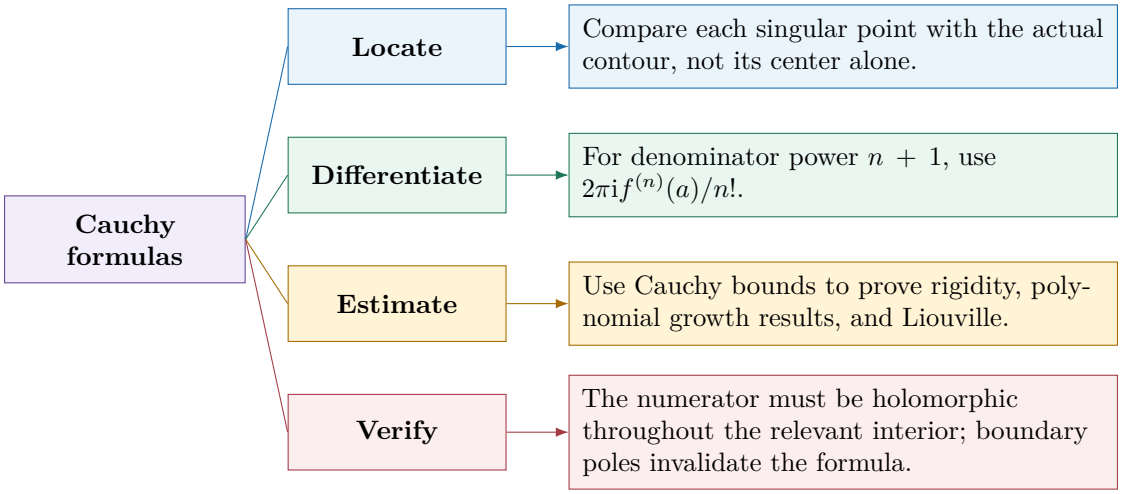


Figure 10.2: Concepts, calculation route, conditions, and checks.

11 Taylor Expansions, Zeros, and the Identity Principle

Source map: Farinati, Chapter 11, pp. 90–97.

11.1 Derive analyticity from the Cauchy kernel

Suppose f is holomorphic near $\overline{D(a,R)}$. On $|w-a|=R$ and $|z-a|\le r<R$,

$$\frac{1}{w-z}=\frac{1}{w-a}\frac{1}{1-(z-a)/(w-a)}=\sum_{n\ge 0}\frac{(z-a)^n}{(w-a)^{n+1}}.$$

The geometric ratio has modulus at most $r/R<1$, uniformly on the contour and the smaller disk. Multiply by the bounded $f(w)$ and integrate term by term in Cauchy's formula:

$$f(z)=\sum_{n\ge 0}a_n(z-a)^n,\qquad a_n=\frac{1}{2\pi i}\oint\frac{f(w)}{(w-a)^{n+1}}\mathrm{d}w=\frac{f^{(n)}(a)}{n!}.$$

The remainder after degree N is at most $M_R(r/R)^{N+1}/(1-r/R)$. This proves a convergent local representation, not merely a formal Taylor polynomial. Termwise differentiation inside the radius shows f is C^∞. Every harmonic function is also smooth locally because it has a local harmonic conjugate.

Worked Example: A shifted expansion and the nearest genuine singularity

Expand $f(z)=1/(1+z^2)$ at $a=1$. Set $w=z-1$. Partial fractions give

$$f(z)=\frac{1}{2i}\left(\frac{1}{z-i}-\frac{1}{z+i}\right),$$

so

$$f(1+w)=\frac{1}{2i}\sum_{n\ge 0}(-1)^n w^n\left[\frac{1}{(1-i)^{n+1}}-\frac{1}{(1+i)^{n+1}}\right].$$

Both geometric expansions require $|w|<\sqrt{2}$. The poles $\pm i$ lie at exactly that distance; a larger Taylor disk would make the represented function holomorphic through a pole, which is impossible. A canceled denominator zero would not impose this obstruction.

11.2 Factor zeros by the first nonzero coefficient

If $f(a) = 0$ and its Taylor series is not identically zero, let $m \geq 1$ be the first index with $a_m \neq 0$. Then

$$f(z) = (z-a)^m g(z), \qquad g(a) = a_m \neq 0.$$

Continuity makes g nonzero near a, so a is an isolated zero of order m. Equivalently $f(a) = \cdots = f^{(m-1)}(a) = 0$ and $f^{(m)}(a) \neq 0$. Orders add under multiplication and subtract under division, provided the residual factors are nonzero.

Worked Example: A cancellation that changes the apparent order

Since $e^z - 1 - z = z^2/2 + z^3/6 + \cdots$, the numerator has a zero of order two. Dividing by z^2 gives a removable singularity with extension value $1/2$ and derivative $1/6$. Dividing instead by z^3 gives a simple pole with residue $1/2$. A denominator exponent alone is not the pole order.

11.3 Identity principle: the accumulation point must be inside

If distinct zeros accumulate at $a \in \Omega$, a finite-order factorization at a is impossible, so all Taylor coefficients vanish and $f = 0$ on a disk. To propagate this through a connected domain, let S be the set of points with a neighborhood on which $f = 0$. It is open. If $a_j \in S$ converge to $a \in \Omega$, all derivatives of f vanish at a_j, and their continuity gives $f^{(k)}(a) = 0$ for every k. Taylor expansion makes $a \in S$. Thus S is relatively closed as well, so $S = \Omega$.

Apply this to $f - g$: two holomorphic functions on a connected domain agreeing on a set with an interior accumulation point agree everywhere. Equality at a single point is not enough. Nor is accumulation only on the boundary.

Worked Example: The interior/boundary distinction

If two entire functions agree at $1/n$ for all positive integers, they agree everywhere, since $1/n \to 0 \in \mathbb{C}$. But on $\mathbb{D}$, $h(z) = \sin(\pi/(1-z))$ is nonzero as a function and vanishes at $z = 1 - 1/n$ for $n \geq 2$. These zeros accumulate at the boundary point 1, not inside the domain.

11.4 Weighted zero sums and finite counting

If f is holomorphic near the compact closure of a bounded Jordan region, is not identically zero, and has no boundary zeros, its interior zeros are finite. Otherwise compactness supplies an accumulation point in the closure; continuity excludes the boundary and the identity theorem excludes an interior one.

At a zero a_j of order m_j, factorization gives $f'/f = m_j/(z-a_j) + g_j'/g_j$. For a holomorphic weight h,

$$\frac{1}{2\pi \mathrm{i}} \oint h(z) \frac{f'(z)}{f(z)} \mathrm{d}z = \sum_j m_j h(a_j).$$

Indeed shrink the contour to small circles about the zeros. The holomorphic terms integrate to zero and Cauchy's formula evaluates each remaining term. This gives the argument principle for holomorphic functions before meromorphic generalization in Chapter 13.

Worked Example: Count and locate a unique zero by two integrals

If a region contains exactly one simple zero a of f and no boundary zeros, then $\oint f'/f = 2\pi\mathrm{i}$ and $\oint z f'/f = 2\pi\mathrm{i}a$. Thus their quotient gives a. If instead the zero has multiplicity m, both integrals acquire the factor m and the quotient still gives its location; if there are several zeros it gives their multiplicity-weighted average, not an individual root.

11.5 Local normal form and open mapping

For nonconstant f, write $f(z) - f(a) = (z-a)^m g(z)$ near a. Choose a holomorphic mth root q of the nonzero g on a small disk. Set $\zeta = (z-a)q(z)$; its derivative at a is $q(a) \neq 0$, so it is a local coordinate. In that coordinate

$$f(z) = f(a) + \zeta^m.$$

A small disk in the ζ-plane maps onto a disk about $f(a)$, proving that every nonconstant holomorphic map is open, even when $f'(a) = 0$. If $m \geq 2$, distinct roots of the same small value show local noninjectivity. Consequently an injective holomorphic map has nonzero derivative everywhere and a holomorphic inverse on its image.

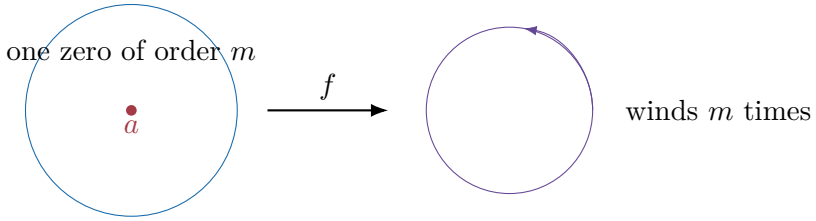


Figure 11.1: A zero of order m gives a local m-to-one map and multiplies winding numbers by m.

Standard Workflow: Taylor and zero questions

Choose the actual center; expand known functions or the Cauchy kernel; state a justified radius; locate the first nonzero coefficient; factor the zero before dividing; test connectedness and the position of an accumulation point before using identity; count multiplicities in weighted zero integrals.

11.6 Chapter Mind Map

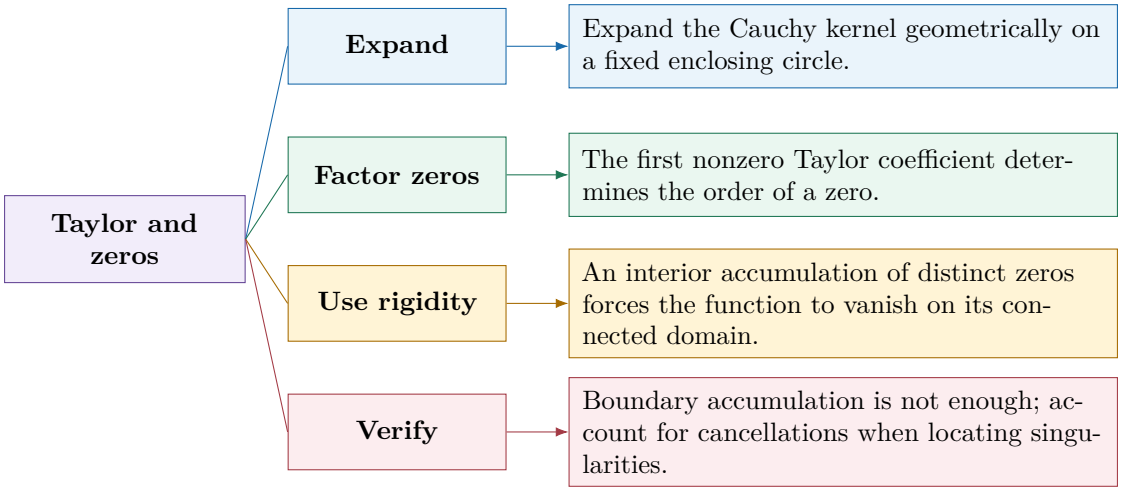


Figure 11.2: Concepts, calculation route, conditions, and checks.

12 Laurent Series, Singularities, and Residues

Source map: Farinati, Chapter 12, pp. 97–116.

12.1 An annulus needs two geometric expansions

Suppose f is holomorphic on $r < |z-a| < R$. Choose radii $r < \rho_1 < |z-a| < \rho_2 < R$. Cauchy's formula on the annulus gives

$$f(z) = \frac{1}{2\pi \mathrm{i}} \oint_{|w-a|=\rho_2} \frac{f(w)}{w-z}\mathrm{d}w - \frac{1}{2\pi \mathrm{i}} \oint_{|w-a|=\rho_1} \frac{f(w)}{w-z}\mathrm{d}w.$$

For the outer circle expand in $(z-a)/(w-a)$; for the inner one expand instead in $(w-a)/(z-a)$. In the latter, $1/(w-z) = -(z-a)^{-1}\sum_{n\geq 0}[(w-a)/(z-a)]^n$, so the leading minus sign cancels the inner-boundary minus. This yields

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-a)^n, \qquad a_n = \frac{1}{2\pi \mathrm{i}} \oint_{|w-a|=\rho} \frac{f(w)}{(w-a)^{n+1}}\mathrm{d}w,$$

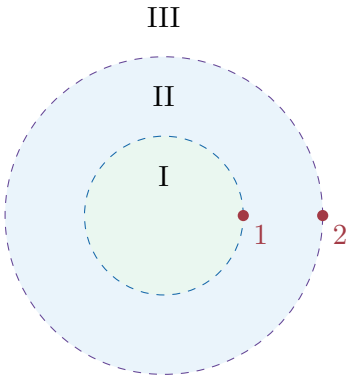
for any $r < \rho < R$. Cauchy's theorem shows the coefficient is independent of ρ. The two series converge uniformly on compact subannuli; integrating termwise proves uniqueness of all coefficients.

Worked Example: The same rational function in three annuli

For $f(z) = 1/[(z-1)(z-2)] = -1/(z-1) + 1/(z-2)$:

$$\begin{aligned} |z|<1: &\quad f = \sum_{n\geq 0}(1-2^{-n-1})z^n, \\ 1<|z|<2: &\quad f = -\sum_{n\geq 0} z^{-n-1} - \sum_{n\geq 0}\frac{z^n}{2^{n+1}}, \\ |z|>2: &\quad f = \sum_{n\geq 0}(2^n-1)z^{-n-1}. \end{aligned}$$

Every geometric ratio is strictly less than one in its stated region. The middle coefficient a_{-1} is -1, while the outer one is zero. There is no contradiction: these are different annuli surrounding different collections of singularities.



different Laurent expansions

I: $|z| < 1$

II: $1 < |z| < 2$

III: $|z| > 2$

Figure 12.1: Distances from the center to singularities determine which geometric expansions are available.

12.2 Classify only isolated singularities

An isolated singularity at a means f is holomorphic on some full punctured disk $0 < |z-a| < R$. Its Laurent principal part gives three mutually exclusive cases:

Key Result: The trichotomy

No negative coefficients: a removable singularity, with extension value a_0. Finitely many negative coefficients, with lowest nonzero power $-m$: a pole of order m, so $f = (z-a)^{-m}g$ with $g(a) \neq 0$. Infinitely many nonzero negative coefficients: an essential isolated singularity.

Boundedness criterion. If $|f| \leq M$ near a, the coefficient formula gives $|a_{-m}| \leq M\rho^m$ for every small ρ, so $a_{-m} = 0$ for $m \geq 1$. Thus boundedness implies removability; continuity of an extension proves the converse.

Pole criterion. A pole has $|f(z)| \to \infty$. Conversely, if this limit holds, $1/f$ is bounded near a and extends with value zero. Its nonzero holomorphic extension has a zero of some finite order m, so reciprocating gives a pole of that order.

Worked Example: Compare four expressions at zero

$\sin z/z$ is removable with value one. $(1-\cos z)/z^4$ has leading term $1/(2z^2)$ and is a pole of order two, residue zero. $\mathrm{e}^{1/z}$ has infinitely many negative powers and is essential, residue one. Finally $1/\sin(\pi/z)$ has poles at $z = 1/n$, $n \in \mathbb{Z} \setminus \{0\}$, accumulating at zero. Thus zero is *not isolated* for this reciprocal, and the isolated-essential classification cannot be applied to it.

12.3 Essential singularities have dense local images

If an essential singularity omitted an entire disk about some value b in a punctured neighborhood, then $h = 1/(f-b)$ would be bounded and extend holomorphically at a. If $h(a) \neq 0$, reciprocating would extend f holomorphically. If $h(a) = 0$, its finite zero order would make $f = b + 1/h$ a pole. Both contradict essentiality. Therefore every punctured neighborhood has image dense in $\mathbb{C}$: this is Casorati–Weierstrass. Density is weaker than attaining every value; $\mathrm{e}^{1/z}$ never equals zero.

12.4 Residues select exactly one coefficient

Define $\operatorname{Res}(f,a) = a_{-1}$. Uniform integration of the Laurent series on a small circle gives

$$\oint f(z)\mathrm{d}z = 2\pi\mathrm{i}a_{-1},$$

because every integer power except -1 has a primitive on the punctured plane. Removing disjoint small disks about finitely many isolated singularities in a Jordan region and applying Cauchy's theorem gives

$$\oint_\gamma f\mathrm{d}z = 2\pi\mathrm{i} \sum_{a_j \text{ inside}} \operatorname{Res}(f,a_j).$$

For an admissible nonsimple cycle, each residue is multiplied by $\operatorname{Ind}(\gamma, a_j)$. No singularity may lie on the path itself.

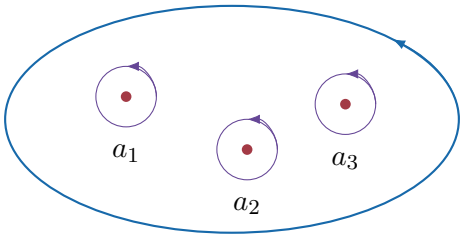


Figure 12.2: Deformation plus local Laurent coefficients yields the residue theorem.

12.5 Choose the simplest residue calculation

If q has a simple zero at a and p is holomorphic, then

$$\operatorname{Res}(p/q, a) = \frac{p(a)}{q'(a)}.$$

This follows from $q = (z-a)q_1$, $q_1(a) = q'(a) \neq 0$. It remains valid with a zero numerator, giving zero, though the quotient may then be removable. At a pole of order m,

$$\operatorname{Res}(f, a) = \frac{1}{(m-1)!}\left[\frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}}((z-a)^m f(z))\right]_{z=a}.$$

This is the Taylor coefficient of degree $m-1$ of the regularized factor. At essential points use coefficient extraction rather than a finite pole formula.

Worked Example: A repeated pole requires the derivative of the whole factor

For $f(z) = \mathrm{e}^z/[z^2(z-1)]$, zero is a double pole. Its residue is

$$\left(\frac{\mathrm{e}^z}{z-1}\right)'_{z=0} = \left(\frac{\mathrm{e}^z}{z-1} - \frac{\mathrm{e}^z}{(z-1)^2}\right)_{z=0} = -2.$$

At one the residue is e. Thus $\oint_{|z|=2} f\mathrm{d}z = 2\pi\mathrm{i}(\mathrm{e}-2)$. Differentiating only the exponential would miss half the residue at zero.

Worked Example: An essential-point residue without classifying it as a pole

For $\mathrm{e}^{1/z}/(1+z^2)$ near zero, expand $\sum z^{-n}/n!$ and $\sum(-1)^k z^{2k}$. A power -1 occurs when $n = 2k+1$, so

$$\operatorname{Res}\left(\frac{\mathrm{e}^{1/z}}{1+z^2}, 0\right) = \sum_{k\geq 0}\frac{(-1)^k}{(2k+1)!} = \sin 1.$$

The products are absolutely convergent on any fixed circle $0 < r < 1$, so this coefficient summation is justified. Chapter 13 obtains the same result from the residue at infinity.

Standard Workflow: Laurent and singularity work

List distances from the expansion center to genuine singularities. Pick the annulus before expanding. Factor cancellations and find the first nonzero local powers. Verify isolation before using the trichotomy. For residues choose simple-zero, repeated-pole, or coefficient extraction as appropriate. Sum only the enclosed residues with the correct winding and orientation.

12.6 Chapter Mind Map

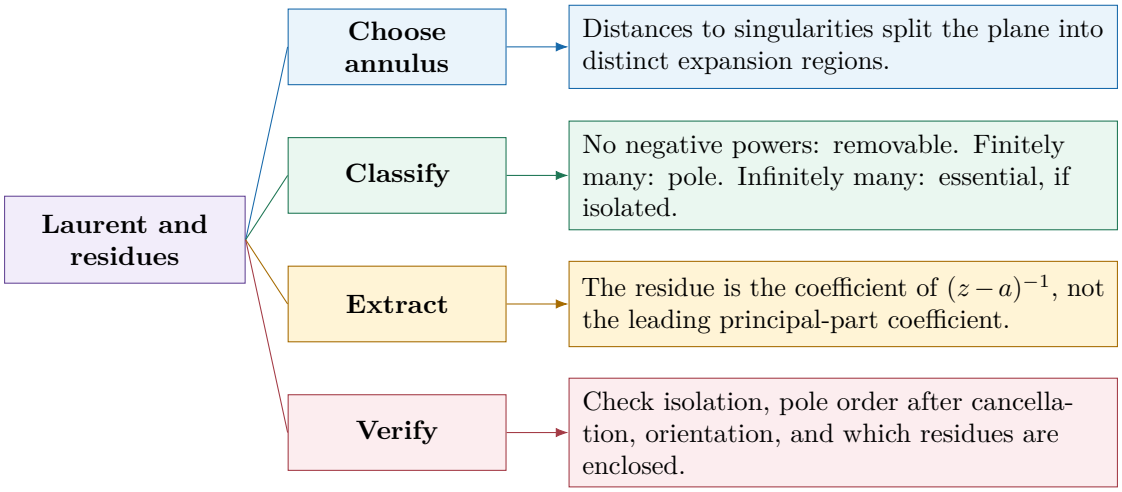


Figure 12.3: Concepts, calculation route, conditions, and checks.

13 Argument Principle, Rouché, and Infinity

Source map: Farinati, Chapter 13, pp. 116–122; weighted zero exercises, pp. 96–97.

13.1 A logarithmic derivative counts multiplicity

For a meromorphic function, locally write $f = (z-a)^m g$ with $g(a) \neq 0$ and integer m; $m > 0$ denotes a zero and $m < 0$ a pole. Then

$$\frac{f'}{f} = \frac{m}{z-a} + \frac{g'}{g}, \qquad \operatorname{Res}(f'/f, a) = m.$$

If f has no zeros or poles on the positive Jordan contour and is meromorphic near its closure, the residue theorem gives

$$\frac{1}{2\pi \mathrm{i}} \oint \frac{f'}{f} \mathrm{d}z = N - P,$$

with multiplicities. The left side is also the winding number of the image curve $f \circ \gamma$ around zero. A small change in f that keeps its boundary image away from zero cannot change this integer.

Worked Example: Zeros minus poles, not just zeros

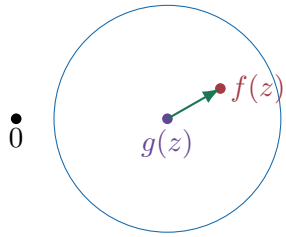
For $f(z) = (z-1)^3(z+2)/(z-\mathrm{i})^2$ on $|z| = 3/2$, the zero at one contributes 3, the zero at -2 is outside, and the pole at i contributes -2. Hence $(2\pi\mathrm{i})^{-1} \oint f'/f = 1$. The function still has three zeros counted with multiplicity inside; the integral is their number minus the two poles.

13.2 Rouché's theorem and a proof by a nonvanishing homotopy

Let g, h be holomorphic near a closed Jordan region and suppose $|h| < |g|$ on its boundary. Then g and $g+h$ have the same number of zeros inside, counted with multiplicity. Indeed $f_t = g + th$ never vanishes on the boundary for $0 \leq t \leq 1$, since $|f_t| \geq |g| - |h| > 0$. Compactness gives a uniform positive lower bound. Therefore

$$N(t) = \frac{1}{2\pi \mathrm{i}} \oint \frac{g' + th'}{g + th} \mathrm{d}z$$

depends continuously on t. It is integer valued by the argument principle, so is constant. At $t = 0$ and one this gives the conclusion. Strict inequality also proves that neither endpoint function has a boundary zero.



$|f-g|<|g|$: the joining segment avoids 0

Figure 13.1: At each boundary point the straight joining segment stays inside a disk that excludes zero.

Worked Example: Two contours reveal an annular root count

Let $p(z)=z^5+3z+1$. On $|z|=1$, choose $g=3z$ and $h=z^5+1$: $|h|\le 2<3=|g|$, so there is one zero in the unit disk. On $|z|=2$, choose $g=z^5$: $|3z+1|\le 7<32$, so all five zeros lie inside radius two. Thus four zeros lie in $1<|z|<2$. No individual quintic root was computed, and none lies on either circle.

Worked Example: A parameter range requires a uniform boundary bound

For $p_a(z)=z^4+az+2$ on $|z|=1$, the constant term dominates if $1+|a|<2$, i.e. $|a|<1$. Then there are no zeros in the unit disk. At $|a|=1$ this boundary estimate is not strict, so Rouché in this form gives no verdict. Indeed $z=\mathrm{e}^{\mathrm{i}\pi/4}$ and $a=-\mathrm{e}^{-\mathrm{i}\pi/4}$ give $p_a(z)=-1-1+2=0$ on the contour. For every $|a|\le 1$, however, $|z^4+az|\le |z|^4+|z|<2$ at interior points, proving there are no interior zeros even in the equality case. Boundary exclusion and interior zero-freeness are distinct conclusions.

13.3 Infinity: transform the differential, not only the function

The type of f at infinity is the type of $f(1/w)$ at zero. A polynomial of degree m has a pole of order m at infinity. A rational function with numerator/denominator degrees p,q has an infinity pole if $p>q$, a zero if $p<q$, and a finite nonzero limit if $p=q$, after cancellations.

For residues the coordinate change includes the Jacobian:

$$f(z)\mathrm{d}z=-\frac{f(1/w)}{w^2}\mathrm{d}w,\qquad \operatorname{Res}(f,\infty)=-\operatorname{Res}\left(\frac{f(1/w)}{w^2},0\right).$$

If $f(z)=\sum a_n z^n$ on a sufficiently large annulus, then $\operatorname{Res}(f,\infty)=-a_{-1}$. A large positive contour surrounds every finite singularity; its inversion is clockwise. Consequently

$$\sum_{a\in\mathbb{C}}\operatorname{Res}(f,a)+\operatorname{Res}(f,\infty)=0$$

when the finite isolated singularities are finite in number and f is holomorphic outside them. They need not all be poles for this contour identity to hold.

Worked Example: A complicated coefficient obtained from easy residues

For $f=\mathrm{e}^{1/z}/(1+z^2)$, the transformed differential factor is $-\mathrm{e}^w/(1+w^2)$, holomorphic at zero, so $\operatorname{Res}(f,\infty)=0$. The finite simple-pole residues are $\mathrm{e}^{-\mathrm{i}}/(2\mathrm{i})$ at i and $\mathrm{e}^{\mathrm{i}}/(-2\mathrm{i})$ at $-\mathrm{i}$. Their sum is $-\sin 1$. Therefore the residue at zero is $\sin 1$, agreeing with the Laurent calculation. The minus sign at infinity is indispensable.

13.4 Reciprocal root sums without solving the polynomial

For $P(z) = a_0 + a_1 z + a_2 z^2 + \cdots$, $a_0 \neq 0$, factor it over $\mathbb{C}$ and differentiate logarithmically:

$$\frac{P'}{P} = \sum_j \frac{m_j}{z - z_j}.$$

Evaluating at zero and differentiating once gives

$$\sum_j \frac{m_j}{z_j} = -\frac{a_1}{a_0}, \qquad \sum_j \frac{m_j}{z_j^2} = \left(\frac{a_1}{a_0}\right)^2 - \frac{2a_2}{a_0}.$$

Equivalently integrate $P'/(zP)$ or $P'/(z^2P)$ on a large circle and include the residue at zero. The coefficient condition $a_0 \neq 0$ ensures no reciprocal root is undefined.

Worked Example: A cubic summarized by its coefficients

For $P(z) = 2 + 3z + 5z^2 + z^3$, the reciprocal root sum is $-3/2$, and the reciprocal-square sum is $9/4 - 5 = -11/4$. A negative or nonreal-looking result is not suspicious: the roots are complex and reciprocal squares are not nonnegative real quantities.

Standard Workflow: Counting and infinity

Identify zeros/poles with their orders; rule out boundary zeros; select a dominant comparison and prove a strict inequality on the whole contour; subtract counts to get annular counts. At infinity use $w = 1/z$ and transform $f\mathrm{d}z$. For root sums inspect the low coefficients before attempting any explicit root formula.

13.5 Chapter Mind Map

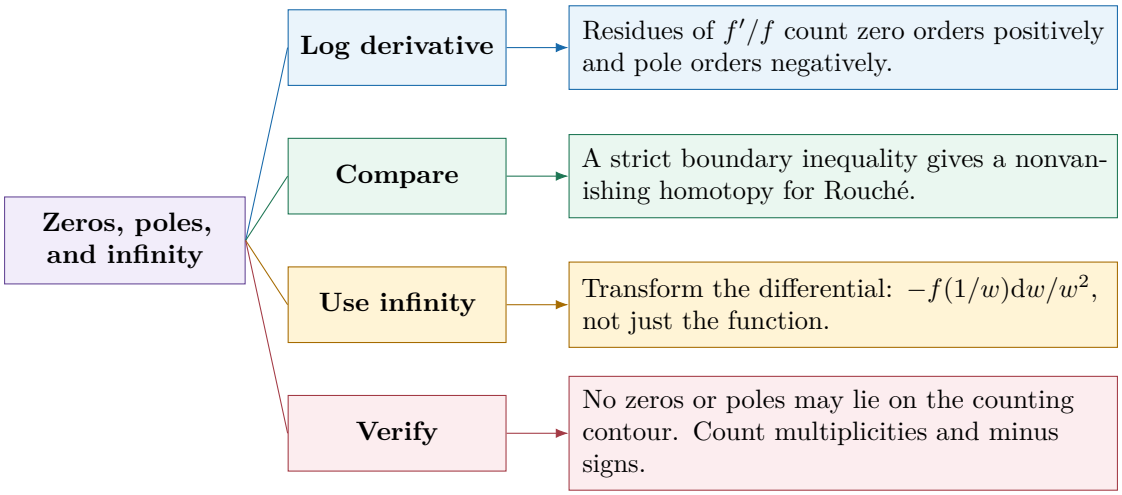


Figure 13.2: Concepts, calculation route, conditions, and checks.

14 Real Integrals by Contour Methods

Source map: Farinati, Chapter 14, pp. 123–132; indentation added as a labeled standard extension.

14.1 Periodic trigonometric integrals

For $z = \mathrm{e}^{\mathrm{i}t}$ on the positive unit circle,

$$\cos t = \frac{z + z^{-1}}{2}, \quad \sin t = \frac{z - z^{-1}}{2\mathrm{i}}, \quad \mathrm{d}t = \frac{\mathrm{d}z}{\mathrm{i}z}.$$

Substitute all three relations and simplify before locating poles. Multiplying to remove negative powers may introduce or cancel a zero at zero; inspect the simplified rational function.

Worked Example: A trigonometric rational integral, fully reduced

$$I = \int_0^{2\pi} \frac{\mathrm{d}t}{5 + 4\cos t} = \frac{1}{\mathrm{i}} \oint_{|z|=1} \frac{\mathrm{d}z}{2z^2 + 5z + 2}.$$

The roots are $-1/2, -2$; only $-1/2$ is inside. The residue of $1/(2z^2+5z+2)$ there is $1/(4(-1/2)+5) = 1/3$. Thus $I = (1/\mathrm{i})(2\pi\mathrm{i})/3 = 2\pi/3$. Positivity is an immediate check. The same expression with $\cos^2 t$ instead of $\cos t$ would be a different integral.

For real $a > |b|$, the same quadratic computation gives $\int_0^{2\pi} \mathrm{d}t/(a+b\cos t) = 2\pi/\sqrt{a^2 - b^2}$. The inequality keeps the denominator positive and the poles off the circle; it is not an optional footnote.

14.2 Rational real-line integrals and a proved arc estimate

Let $R = P/Q$ have no real poles after cancellation, with $\deg Q \geq \deg P + 2$. For a large upper semicircle, $|R(z)| \leq C/|z|^2$, hence

$$\left| \int_{\mathrm{arc}} R(z)\mathrm{d}z \right| \leq \pi R \frac{C}{R^2} \to 0.$$

The straight segment tends to the absolutely convergent real integral. Therefore

$$\int_{-\infty}^{\infty} R(x)\mathrm{d}x = 2\pi\mathrm{i} \sum_{\mathrm{Im}\, a > 0} \mathrm{Res}(R, a).$$

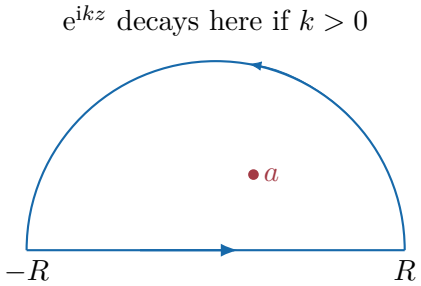


Figure 14.1: The real segment and upper arc form a positive contour. Decay must be proved on the arc, not only on the real axis.

Worked Example: A quartic with two upper poles

For $R(z) = 1/(1 + z^4)$, the upper poles are $a = \mathrm{e}^{\mathrm{i}\pi/4}$ and $b = \mathrm{e}^{3\pi\mathrm{i}/4}$. Each residue is $1/(4z^3) = -z/4$ because $z^4 = -1$. Their sum is $-(a + b)/4 = -\mathrm{i}\sqrt{2}/4$. Thus

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{1 + x^4} = \frac{\pi}{\sqrt{2}}.$$

Worked Example (continued)

The half-line integral is half this value by evenness. Keeping only one upper pole would halve the real-line answer incorrectly.

Worked Example: Coalescing poles require their own case

For $a, b > 0$, $a \neq b$,

$$\int_{\mathbb{R}} \frac{\mathrm{d}x}{(x^2+a^2)(x^2+b^2)} = 2\pi\mathrm{i}\left[\frac{1}{2\mathrm{i}a(b^2-a^2)} + \frac{1}{2\mathrm{i}b(a^2-b^2)}\right] = \frac{\pi}{ab(a+b)}.$$

At $a = b$, do not use the two simple-pole expressions separately. The double pole at $\mathrm{i}a$ has residue $[-2/(z+\mathrm{i}a)^3]_{z=\mathrm{i}a} = 1/(4\mathrm{i}a^3)$, giving $\pi/(2a^3)$, which agrees with the limit of the simplified formula.

14.3 Fourier factors and Jordan's lemma

Assume the rational factor has no real poles after cancellation. A pole on the real axis needs a separate indentation or principal-value analysis, as discussed below. For real $k > 0$, $|\mathrm{e}^{\mathrm{i}kz}| = \mathrm{e}^{-k\,\mathrm{Im}\,z}$ decays in the upper half-plane. The cosine alone contains both a growing and a decaying exponential there; integrate the single exponential and take real or imaginary parts at the end.

For $0 \le t \le \pi/2$, concavity gives $\sin t \ge 2t/\pi$. Hence

$$\int_0^{\pi} \mathrm{e}^{-kR\sin t}\mathrm{d}t \le 2\int_0^{\pi/2} \mathrm{e}^{-2kRt/\pi}\mathrm{d}t \le \frac{\pi}{kR}.$$

If $M_R = \sup_{\mathrm{arc}} |R(z)| \to 0$, the exponential-weighted arc integral is bounded by $RM_R\pi/(kR) = \pi M_R/k \to 0$. In particular rational functions of degree difference at least one work. The estimate is for $k > 0$; $k = 0$ needs ordinary rational decay. For $k < 0$, close below and multiply the lower residue sum by $-2\pi\mathrm{i}$ because the closure is clockwise.

Worked Example: A transform and its sign-symmetric answer

For $a > 0$, $k > 0$, the only upper pole of $\mathrm{e}^{\mathrm{i}kz}/(z^2+a^2)$ is $\mathrm{i}a$, residue $\mathrm{e}^{-ak}/(2\mathrm{i}a)$. Thus

$$\int_{\mathbb{R}} \frac{\mathrm{e}^{\mathrm{i}kx}}{x^2+a^2}\mathrm{d}x = \frac{\pi}{a}\mathrm{e}^{-a|k|} \quad (k \in \mathbb{R}).$$

The lower contour gives the stated value for $k < 0$, and $k = 0$ follows from a direct rational contour. The imaginary part is zero by oddness; the real part is the cosine transform. This is a direct bridge to wave/Fourier and probability calculations.

14.4 A boundary pole is not automatically half a residue

Supplementary Background: Principal values and indentation

For a simple real pole, an upper small semicircle traversed left to right is clockwise. If the residue is r, its contribution tends to $-\pi\mathrm{i}r$, because $\int r\mathrm{d}z/(z-a)$ changes its argument by $-\pi$. The rest of the contour must still be evaluated. This is a statement about an indented contour and a Cauchy principal value, not an ordinary integral through a pole.

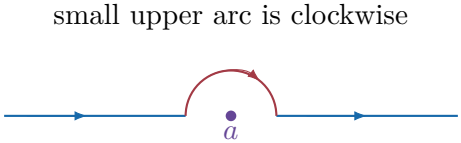


Figure 14.2: The orientation fixes the sign of a small indentation contribution.

Worked Example: Why the sinc integral is finite

Integrate e^{iz}/z over the upper semicircle with a small upper indentation about zero. The pole is excluded, so the closed contour encloses no singularity. The large arc vanishes by Jordan's estimate, and the small arc tends to $-\pi i$. Hence PV $\int_{\mathbb{R}} e^{ix} dx/x = \pi i$. Taking imaginary parts gives

$$\int_{\mathbb{R}} \frac{\sin x}{x} dx = \pi.$$

Here the real function $\sin x/x$ has a removable value one at zero and converges conditionally at infinity by integration by parts. Its integral is ordinary improper, whereas the complex e^{ix}/x required a principal value at zero. The half-line sinc integral is $\pi/2$.

14.5 A rectangle where a semicircle would fail

Let $I(k) = \int_{\mathbb{R}} e^{-x^2} e^{-ikx} dx$. Completing the square gives

$$-x^2 - ikx = -(x + ik/2)^2 - k^2/4.$$

For $b = k/2$, integrate e^{-z^2} around the rectangle with horizontal sides at imaginary heights zero and b. A vertical side at $\operatorname{Re} z = \pm R$ has modulus bounded by $e^{-R^2+b^2}$ and length $|b|$, so its integral vanishes. The top side has the opposite traversal direction, giving

$$\int_{\mathbb{R}} e^{-(x+ib)^2} dx = \int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}.$$

The Gaussian value follows by squaring the positive integral and using polar coordinates: $I(0)^2 = 2\pi \int_0^\infty r e^{-r^2} dr = \pi$. Thus

$$I(k) = \sqrt{\pi} e^{-k^2/4}.$$

The side bound uses $|b|$ and works for negative k as well. A large semicircle is inappropriate because e^{-z^2} grows along imaginary directions.

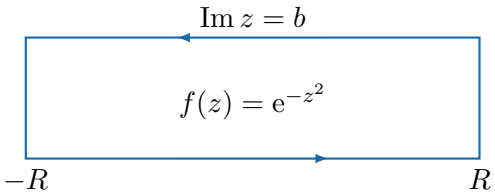


Figure 14.3: A bounded vertical shift controls the Gaussian where a semicircle would encounter exponential growth.

Standard Workflow: Real integral workflow

Check convergence and parameters first. Choose a substitution or contour from the integrand's structure. Locate every pole, prove the large-arc or side estimate, treat boundary singularities separately, compute residues, and only then take real/imaginary parts. Finish with parity, reality, positivity, dimensions if applicable, and a parameter-limit check.

14.6 Chapter Mind Map

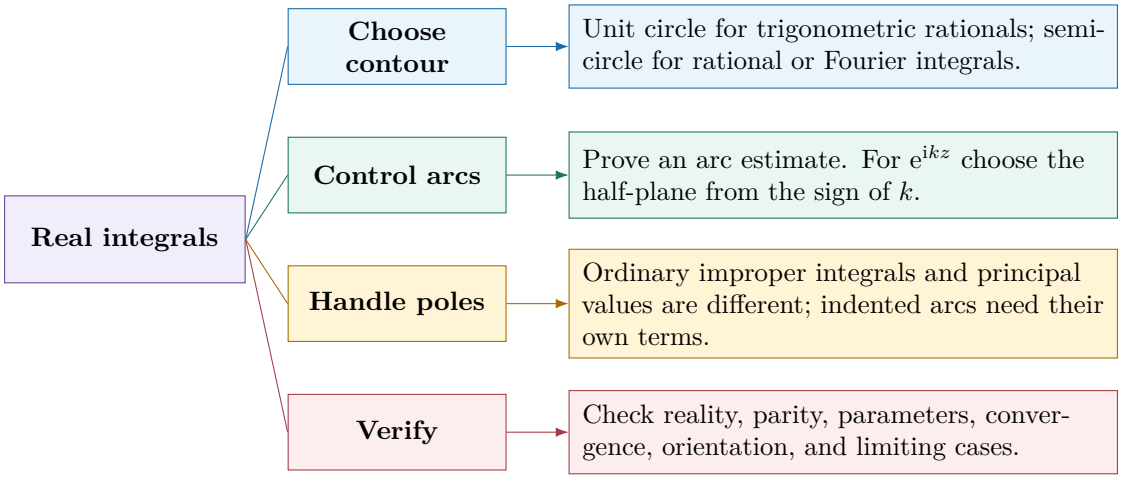


Figure 14.4: Concepts, calculation route, conditions, and checks.

15 Optional Enrichment: Disk Maps

Source map: Farinati, Chapter 15, pp. 133–142; compactness details and Schwarz–Pick added for explanation.

Error Check: Scope status

The current syllabus explicitly requires conformal mappings, covered already in Chapters 5–6 and the elementary constructions below. It does not explicitly name Schwarz's lemma, disk-automorphism classification, normal families, or the Riemann mapping theorem. These lecture-Chapter-15 results are retained as optional enrichment, not presented as verified examination requirements. Complete the core Chapters 1–14 first.

15.1 Schwarz lemma and every equality case

Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic with $f(0) = 0$. Define $g(z) = f(z)/z$ for $z \neq 0$, and $g(0) = f'(0)$ by the removable Taylor expansion. On $|z| = r < 1$, $|g| \leq 1/r$. Maximum modulus bounds g inside this circle; for fixed z, let $r \uparrow 1$ to obtain

$$|f(z)| \leq |z|, \qquad |f'(0)| \leq 1.$$

If $|f(a)| = |a|$ at some $a \neq 0$, or $|f'(0)| = 1$, then $|g|$ attains its upper bound one inside, so $g \equiv e^{i\theta}$ and $f(z) = e^{i\theta} z$. Conversely these rotations attain both equalities.

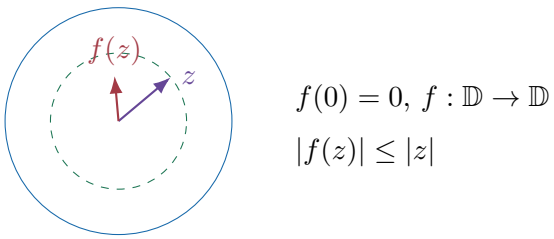


Figure 15.1: A disk self-map fixing zero cannot increase distance from zero; equality forces a rotation.

15.2 All holomorphic bijections of the disk

For $a \in \mathbb{D}$, define

$$\phi_a(z) = \frac{z-a}{1-\bar{a}z}, \qquad \phi_a'(z) = \frac{1-|a|^2}{(1-\bar{a}z)^2}, \qquad \phi_a^{-1} = \phi_{-a}.$$

The denominator cannot vanish in $\mathbb{D}$. Expanding squared moduli gives the essential identity

$$1-|\phi_a(z)|^2 = \frac{(1-|a|^2)(1-|z|^2)}{|1-\bar{a}z|^2} > 0.$$

Thus ϕ_a maps the disk into itself; the inverse formula proves onto-ness. It sends a to zero and zero to $-a$, with $\phi_a'(a) = 1/(1-|a|^2)$ and $\phi_a'(0) = 1-|a|^2$.

If a disk automorphism F fixes zero, apply Schwarz to both F and F^{-1}:

$$|z| = |F^{-1}(F(z))| \le |F(z)| \le |z|.$$

Equality forces a rotation. For an arbitrary automorphism let $a = F^{-1}(0)$; then $F \circ \phi_a^{-1}$ fixes zero, so

$$F(z) = \mathrm{e}^{\mathrm{i}\theta}\frac{z-a}{1-\bar{a}z}.$$

Worked Example: A normalized map without guessing its rotation

Find the disk automorphism with $F(1/2) = 0$ and $F'(1/2) > 0$. Every such map is $\mathrm{e}^{\mathrm{i}\theta}\phi_{1/2}$. Its derivative at $1/2$ is $4\mathrm{e}^{\mathrm{i}\theta}/3$, positive real only when the rotation is one. Hence $F = (z-1/2)/(1-z/2)$ uniquely.

Worked Example: Two fixed interior points force identity

If $F(a) = a$ and $F(b) = b$ with $a \neq b$ in the disk, set $G = \phi_a \circ F \circ \phi_a^{-1}$. It fixes zero, hence $G(w) = \mathrm{e}^{\mathrm{i}\theta}w$. It also fixes $\phi_a(b) \neq 0$, so $\mathrm{e}^{\mathrm{i}\theta} = 1$. Therefore F is the identity. The statement is about interior fixed points of an automorphism; two boundary fixed points need not force identity.

Supplementary Background: Schwarz–Pick: normalization produces the general inequality

For any holomorphic $F : \mathbb{D} \to \mathbb{D}$, normalize at a by $G = \phi_{F(a)} \circ F \circ \phi_a^{-1}$. Schwarz gives

$$\left|\frac{F(z)-F(a)}{1-\overline{F(a)}F(z)}\right| \le \left|\frac{z-a}{1-\bar{a}z}\right|, \qquad \frac{|F'(a)|}{1-|F(a)|^2} \le \frac{1}{1-|a|^2}.$$

Equality in the derivative inequality, or a pair inequality with $z \neq a$, makes the normalized map a rotation and hence F an automorphism. The denominator is never zero for disk points.

15.3 Compose elementary maps with exact image ranges

The Cayley map $(z-\mathrm{i})/(z+\mathrm{i})$ sends $\mathbb{H}$ onto $\mathbb{D}$. A sector $0 < \operatorname{Arg} z < \alpha$, $0 < \alpha < 2\pi$, maps to $\mathbb{H}$ by $z^{\pi/\alpha}$ with that argument branch; then compose with Cayley. The slit plane $\mathbb{C} \setminus (-\infty, 0]$ maps to the right half-plane by its principal square root, then to the disk by $(w-1)/(w+1)$. The strip $0 < \operatorname{Im} z < \pi$ maps bijectively to $\mathbb{H}$ by e^z.

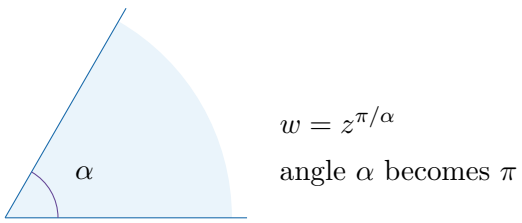


Figure 15.2: A power straightens a corner only after its branch and angular range have been fixed.

Worked Example: A slit-plane bijection and its inverse

$$F(z) = \frac{\sqrt{z}-1}{\sqrt{z}+1}, \qquad F^{-1}(w) = \left(\frac{1+w}{1-w}\right)^2.$$

For $|w| < 1$, the inner fraction has real part $(1-|w|^2)/|1-w|^2 > 0$, so squaring lands in the principal slit plane and the principal square root recovers it. Thus the maps are mutual inverses; $F(1) = 0$, $F'(1) = 1/4 > 0$. Stating the formula without the branch would not specify a map.

15.4 Riemann mapping: statement and nonempty normalized family

Every nonempty simply connected proper domain $\Omega \subsetneq \mathbb{C}$ is biholomorphic to $\mathbb{D}$. Given $z_0 \in \Omega$, there is a unique bijection with $f(z_0) = 0$ and $f'(z_0) > 0$. The exclusions matter: $\mathbb{C}$ cannot map bijectively to the disk by Liouville, and an annulus cannot by preservation of simple connectivity.

Define $\mathcal{F} = \{f : \Omega \to \mathbb{D} : \ f \text{ holomorphic and injective, } f(z_0) = 0\}$. To show it is nonempty, pick $a \notin \Omega$ and a holomorphic square root $h(z) = \sqrt{z-a}$ on the simply connected domain. It is injective, and $h(\Omega)$ contains no opposite pair $w, -w$: their squares would give the same z, contradicting single-valuedness unless $w = 0$, which is impossible.

Choose $r > 0$ with $D(h(z_0), 2r) \subset h(\Omega)$. Then $D(-h(z_0), 2r)$ misses $h(\Omega)$, so $T(w) = r/(w + h(z_0))$ is injective there and has modulus at most $1/2$. Compose $T \circ h$ with the disk automorphism sending its value at z_0 to zero. This constructs an element of $\mathcal{F}$.

15.5 Existence of an extremal: the compactness step

For a small closed disk about z_0 inside Ω, Cauchy's estimate bounds $|f'(z_0)| \le 1/r_0$ uniformly over $\mathcal{F}$. Hence

$$0 < M = \sup_{f \in \mathcal{F}} |f'(z_0)| < \infty.$$

Positivity uses the nonempty family and the nonzero derivative of an injective holomorphic map. Choose f_n approaching this supremum.

Here is the compactness mechanism behind Montel's theorem in this bounded case. On a compact K choose a slightly larger compact neighborhood inside Ω; Cauchy estimates bound derivatives uniformly there, giving uniform local Lipschitz bounds and thus equicontinuity on K. Arzelà–Ascoli on an exhausting sequence of compact sets, followed by a diagonal subsequence, gives locally uniform convergence to f. The real compactness theorem Arzelà–Ascoli is an input; its full general proof is not claimed here. Chapter 8 proves f holomorphic and convergence of derivatives. Since $|f'(z_0)| = M > 0$, the limit is nonconstant; maximum modulus gives $|f| < 1$.

To verify injectivity of this nonconstant limit, suppose $f(a) = f(b)$ with $a \neq b$. Choose a small circle about b excluding a, on which $f - f(a)$ is nonzero. Its zero at b is isolated. Local uniform convergence makes $f_n - f_n(a)$ close enough on this circle for Rouché, so it has a zero inside. But injectivity of f_n says its only zero is a, outside that circle: a contradiction. Thus $f \in \mathcal{F}$ and attains M.

15.6 The square-root trick proves surjectivity

Suppose $b \in \mathbb{D} \setminus f(\Omega)$. Since $f(z_0) = 0$, $b \neq 0$. The nonvanishing holomorphic function $\phi_b \circ f$ on simply connected Ω has a square root q, with $q^2 = \phi_b(f)$. It maps into $\mathbb{D}$ and is injective because equality of its values implies equality of f values. Set $s = q(z_0)$, so $s^2 = -b$ and $|s| = \sqrt{|b|}$. Then $F = \phi_s \circ q$ lies in $\mathcal{F}$. Differentiating $q^2 = \phi_b(f)$ at z_0 gives $2sq'(z_0) = (1 - |b|^2)f'(z_0)$. Therefore

$$|F'(z_0)| = \frac{|q'(z_0)|}{1-|s|^2} = \frac{1+|b|}{2\sqrt{|b|}}|f'(z_0)|.$$

For $0 < |b| < 1$, $(1 - \sqrt{|b|})^2 > 0$ implies this factor is strictly greater than one. This contradicts maximality. Hence no such b exists, and the extremal is onto.

Finally rotate its derivative to make it positive. If normalized f, g both exist, $f \circ g^{-1}$ is a disk automorphism fixing zero, hence a rotation. Its derivative at zero is $f'(z_0)/g'(z_0) > 0$, so the rotation is one and $f = g$. This completes the proof, with its compactness input explicitly identified.

Standard Workflow: Mapping problems

Identify source and target geometry, remove translations/scales, choose branches for powers and logarithms, compose known bijections, and verify inverse ranges. Normalize last using a disk automorphism and a rotation. For an existence question check properness, connectedness, and simple connectivity; Riemann mapping gives existence, not automatically a closed elementary expression.

15.7 Chapter Mind Map

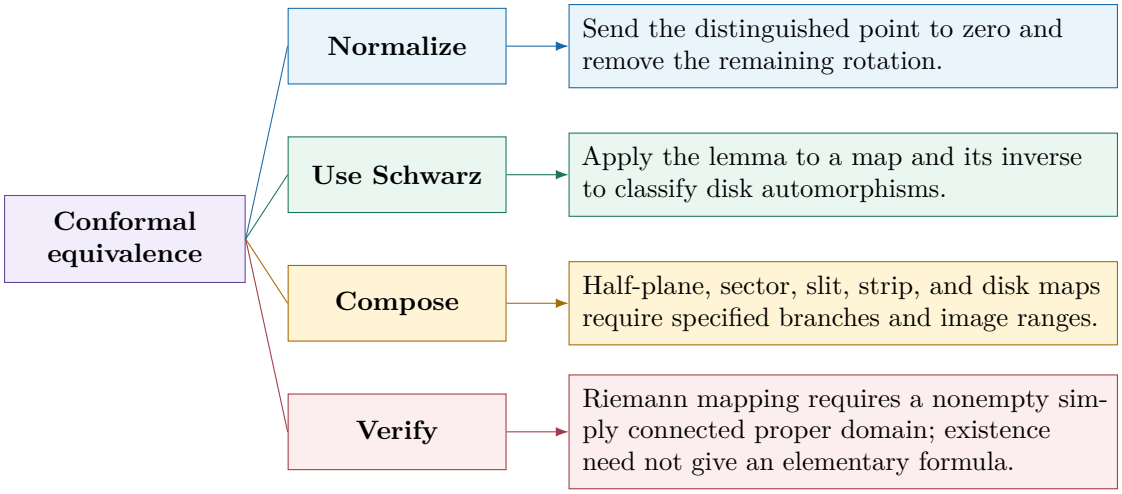


Figure 15.3: Concepts, calculation route, conditions, and checks.

A Formula Index, Method Selection, and Source Checks

A.1 A compact method-selection index

Prompt feature	Standard first move
Complex roots or products	Polar form; list angles modulo 2π before dividing by the root order.
Real/imaginary parts supplied	Check real smoothness, then C–R; integrate a conjugate with an integration function.
Logarithm or noninteger power	State a branch domain and argument interval; inspect loops and periods.
Line/circle image	Möbius pole and three points, then an interior side test.
Infinite series	Radius first; boundary and absolute/conditional convergence second.
Function limit or termwise derivative	Identify the convergence set; use compact uniform bounds and holomorphy.
Open contour	Parameterize or find a global primitive on the actual domain.
Closed contour	Inspect the interior and singularities before choosing Cauchy or residues.

Prompt feature	Standard first move
Entire function with a global bound	Convert the bound to Cauchy estimates or a bounded exponential/reciprocal.
Taylor/Laurent request	Fix the center and all annuli; choose each geometric ratio accordingly.
Unknown zero count	Argument principle or strict Rouché comparison; count multiplicities.
Real integral	Select a unit circle, semicircle, indentation, or rectangle; prove decay.
Conformal bijection	Compose elementary bijections and normalize, or use the precise Riemann mapping hypotheses.

A.2 Reusable formulas with their conditions

$$f' = u_x + \mathrm{i}v_x, \qquad u_x = v_y, \qquad u_y = -v_x,$$

$$T'(z) = \frac{ad-bc}{(cz+d)^2}, \qquad T^{-1}(w) = \frac{dw-b}{-cw+a},$$

$$R^{-1} = \limsup |a_n|^{1/n}, \qquad \operatorname{Ind}(\gamma, a) = \frac{1}{2\pi\mathrm{i}} \oint_\gamma \frac{\mathrm{d}z}{z-a},$$

$$\oint_\gamma \frac{f(z)}{(z-a)^{n+1}}\mathrm{d}z = \frac{2\pi\mathrm{i}}{n!} f^{(n)}(a), \qquad |f^{(n)}(a)| \le n! M_R / R^n,$$

$$\operatorname{Res}(p/q, a) = p(a)/q'(a) \quad (q'(a) \neq 0), \quad \operatorname{Res}(f, \infty) = -\operatorname{Res}(f(1/w)/w^2, 0).$$

The Cauchy integral formula requires an admissible positive contour enclosing a and a holomorphic numerator throughout the relevant filled region. Derivative estimates require the entire radius-R disk to lie in the holomorphy domain. The simple residue formula requires a simple denominator zero. The logarithmic derivative counts zeros minus poles only when the boundary has neither. None of these domain conditions disappears in a formula index.

A.3 Corrections that prevent propagated source errors

The supplied preliminary lecture file contains several apparent typographical or mathematical slips. The following source expressions were checked mathematically and are corrected throughout these notes and practice.

Source location	Correct version used here
pp. 18, 24	$f' = u_x + \mathrm{i}v_x = v_y - \mathrm{i}u_y$; the harmonic cancellation is $(v_y)_x - (v_x)_y = 0$.
p. 27	For $u = \log(x^2+y^2)$ a local conjugate is $2\arg z$, and $f = 2\operatorname{Log} z$, not $\operatorname{Log} z$.
p. 36	For $c \neq 0$, $(az+b)/(cz+d)$ sends infinity to a/c, not a/d.
pp. 78–79	Ordinary Cauchy integral evaluations carry the factor $2\pi\mathrm{i}$, unless the integral has explicitly been normalized by it.
p. 111	$1/\sin(\pi/z)$ has poles at $1/n$; they accumulate at zero, so zero is not an isolated singularity of this reciprocal.
p. 124	The integral computed on pp. 123–124 is the one with $5 + 4\cos t$, not $5 + 4\cos^2 t$; the derivative of $2z^2 + 5z + 2$ is $4z + 5$.
p. 134	ϕ_a has representing matrix $\left(\begin{smallmatrix} 1 & -a \\ -\bar{a} & 1 \end{smallmatrix}\right)$; the lower-left minus sign is required.
p. 135	The principal square root of the slit plane has image the entire right half-plane, with no strip restriction on its imaginary part.